

Supplementary Information

Splicing-factor expression in cultured human fibroblasts reports proliferative state and the coupling weakens in skin

[Authors and affiliations to be completed before submission]

This file contains Supplementary Results 1 to 8, Supplementary Figures S1 to S8, Supplementary Tables 1 to 10, Supplementary Notes 1 and 2 and the list of Supplementary Data files. Reference numbers are those of the main reference list. Every number here is reproduced by the scripts named in the Code availability statement.

Supplementary Results

Supplementary Results 1. Division rate under nine treatments and two oxygen tensions (post hoc)

Division rate was measured in the Cellular Lifespan Study^{13,14} rather than manipulated by us, but nine of its treatment arms manipulate it in the same cell lines. Each treated culture was compared with the control culture of the same cell line and oxygen tension nearest in time in culture (224 of 224 treated cultures matched; median difference in days grown 0). The splicing score is the 177-gene score of Fig. 2. The change a treatment makes to the score follows the change it makes to the measured division rate ($\rho = 0.49$, $P = 4 \times 10^{-15}$, $n = 224$ cultures) and follows the change in the transcriptional proliferation score more closely ($\rho = 0.79$, $P = 1 \times 10^{-49}$; Supplementary Fig. S1a, b). Contact inhibition, the largest manipulation of rate available, lowers the rate by 0.35 divisions per day and the score by 0.73 standard deviations ($n = 41$ cultures).

Pharmacology separates the two measures of proliferation. With the measured division rate held constant, five of the nine arms keep a residual effect on the splicing score at $FDR < 0.05$: dexamethasone +0.53 (95% CI 0.43 to 0.64), oligomycin +0.56 (0.41 to 0.71), mitochondrial nutrient-uptake inhibitors with dexamethasone +0.45 (0.31 to 0.58), oligomycin with dexamethasone +0.41 (0.16 to 0.66) and contact inhibition -0.17 (-0.31 to -0.03); a sixth, galactose, is nominal (+0.15, 95% CI 0.002 to 0.31, $P = 0.048$, $FDR = 0.07$). With the counted rate held constant the four drug treatments among them also carry a higher proliferation score than the rate predicts (dexamethasone +1.27, 1.10 to 1.44; mitochondrial nutrient-uptake inhibitors with dexamethasone +1.08; oligomycin with dexamethasone +0.98; oligomycin +0.54). In the matched differences the proliferation score rises under dexamethasone (+0.81) and under the inhibitors with

dexamethasone (+0.28) and falls under oligomycin (−0.14) and oligomycin with dexamethasone (−0.27). With the proliferation score held constant instead, seven of the nine arms have no residual effect (Supplementary Fig. S1c, Supplementary Table 2); the exceptions are oligomycin, which raises the splicing score above what the proliferation score predicts (+0.28, 0.17 to 0.40), and 2-deoxyglucose, which lowers it (−0.32, −0.43 to −0.20). The proliferation score and the splicing score are strongly correlated with each other, so this second adjustment cannot separate them; what it shows is that these treatments do not move one without the other.

Oxygen tension leaves a further residual. The treatment comparison holds oxygen fixed by design, so it was examined separately among control cultures: fifteen cultures of three cell lines grown at 3% oxygen were matched to control cultures of the same line at 21% oxygen nearest in days grown (median difference 2.3 days). Growth at 3% oxygen lowers the measured division rate by 0.10 divisions per day (12 of 15 cultures lower) and the splicing score by 0.51 (12 of 15 lower), while the proliferation score does not move detectably (−0.26, 10 of 15 lower). In a model of the 104 control libraries with cell line and days grown as covariates, the splicing score is 0.21 lower at 3% oxygen with the measured rate held constant (95% CI −0.39 to −0.03, $P = 0.022$) and 0.33 lower with the proliferation score held constant (−0.44 to −0.23, $P = 2 \times 10^{-8}$). This residual is explained by neither measure of proliferation. It rests on three cell lines, and it counts against a reading in which proliferation is the only determinant of the score.

Apart from contact inhibition and the six cultures given oligomycin with dexamethasone (both −0.35 divisions per day), the arms move the division rate by 0.14 divisions per day or less, so this analysis tests the coupling under perturbation rather than across the full range of division rates. Treatments were not randomised to cell lines and the cultures are serial passages of a few lines. The P values treat cultures as units; the number of cell lines behind each arm (two to seven; Supplementary Table 2a) and, for oxygen, the per-line differences in Supplementary Table 2d are the conservative reading.

Supplementary Results 2. Which genes keep an age association

If division rate drives the association, individual genes should retain their age association in proportion to how loosely they are coupled to division. In the 107 adult donors, 137 of the 177 genes show an age association and every one of them is a decline. Once a transcriptional proliferation score is included, 19 keep a significant decline and none acquires one (Supplementary Fig. S2c). That last statement holds for the primary cohort only: two genes acquire an association after adjustment in the 133-donor definition and eight in the deposited 142-sample set (Supplementary Table

1). The genes are individually well coupled to proliferation (median $r = 0.64$; Supplementary Fig. S2b), and the share of the age effect a gene keeps falls as that coupling rises ($\rho = -0.45$, 95% CI -0.56 to -0.32 , $P = 3 \times 10^{-10}$, $n = 177$ genes). A gene's retained share and its correlation with the covariate being adjusted for are not independent quantities, so this relationship is expected in part from the adjustment itself and is not an independent test. The 19 genes that keep a decline are somewhat less tightly coupled than the rest (median 0.49 against 0.65), although the difference is not significant ($P = 0.16$). None of them is one of the twenty factors the ageing literature assays: of the eleven of those factors that belong to the set, none keeps an age association once proliferation is included (Supplementary Fig. S2a).

The set is not carried by its non-spliceosomal members. Restricted to the 127 genes that belong to Reactome mRNA splicing itself, the age effect and its attenuation are the same as for the whole set (-0.132 per decade, 66% removed by adjustment), and so they are for the remaining 50 genes of capping, 3'-end processing and export (-0.136 , 78% removed; Supplementary Table 3); scored in one matrix, the two subsets follow the measured division rate equally closely ($\rho = 0.66$ and 0.70 , whole set 0.68 in that matrix; see Supplementary Results 6).

Those 19 genes include core spliceosomal and 3'-processing components — *SF3A3*, *SF3B6*, *SNRPF*, *PPIH*, *PPIL1*, *SNW1*, *CDC5L*, *CWC22*, *CWC27*, *MAGO*H, *PNN*, *HNRNPC*, *RBMX*, *PABPN1*, *ZCRB1*, *CCAR1*, *IK*, *THOC2* and *NUP88*. They are not, however, a stable set. The same analysis repeated under the other cohort definitions keeps 40 genes in the deposited 142-sample set, 56 when only progeria donors are removed, 19 in the primary cohort and none among adults aged 20 to 82 (Fig. 4d, Supplementary Table 1). No gene is therefore common to all four definitions; the three definitions that keep any gene share ten (*IK*, *MAGO*H, *PABPN1*, *PNN*, *PPIH*, *PPIL1*, *SF3A3*, *SF3B6*, *SNRPF* and *SNW1*). In the primary specification a decline remains after adjustment at the level of the whole set, but which genes carry it depends on how the cohort is drawn, and no subset of these genes is recommended here as an alternative index of ageing.

Supplementary Results 3. The splicing outcome index carries no age association in the adult cohort

Splicing-factor abundance and the accuracy of splicing are different quantities, and the tissue evidence for age-associated splicing dysfunction rests on the latter⁸. The donor cohort was reprocessed uniformly with junction-level output¹⁷, so both could be taken from the same libraries and treated identically.

The outcome measure does not support an ageing interpretation in this cohort. The share of junction-spanning reads falling on junctions absent

from the reference annotation is only weakly related to proliferation ($r = -0.27$, $n = 107$ donors), and in the 107 adult donors it has no association with age ($\beta = +1.5 \times 10^{-5}$ per decade, 95% CI -3.0×10^{-5} to $+6.0 \times 10^{-5}$, $P = 0.51$), before or after adjustment (Supplementary Fig. S3a, b). The same analysis run on all 142 deposited samples gives an age association ($P = 0.002$) that the cohort definition removes: progeria donors and children contribute to it, and the 31 donors aged 83 and over — every one of them from the same cell repository, which has a higher unannotated-junction fraction at every age — carry the remainder. Among adults aged 20 to 82 the estimate runs in the opposite direction ($P = 0.012$). The index also behaves like a measure of library quality rather than of biology: across these donors it falls with the proportion of uniquely mapped reads ($\rho = -0.43$) and rises with multi-mapping ($\rho = +0.42$) and intronic signal ($\rho = +0.31$), and adding those quality measures to the model removes what little age association is left ($P = 0.96$), while the age effect on splicing-factor expression survives them (-0.110 per decade, $P = 7 \times 10^{-6}$, and -0.039 after adjustment for proliferation, $P = 0.003$; Supplementary Table 3). The entropy of junction usage behaves the same way. This index was not specified in advance of the analyses in which it first appeared and is reported as exploratory.

Event-level splicing carries the same warning (Supplementary Fig. S3c, d). The number of age-associated events falls from 135, 302, 332 and 98 under the four cohort definitions to 20, 22, 14 and none once proliferation is included. The *COL1A2* event with the strongest age association in the deposited set is flat among adults below 83 (PSI 0.035 to 0.049 by age band; $\rho = 0.12$, $P = 0.29$, $n = 76$ adults aged 20 to 82; the $\rho = 0.22$ printed in Supplementary Fig. S3d includes donors under 20) and steps up to 0.072 in the 83-and-over stratum, and the distal junction that produces it is absent from GENCODE v41. Whether splicing outcome deteriorates with age in these cells therefore cannot be decided from this cohort, and the tissue-level evidence for age-associated splicing degeneration⁸ stands on its own measurements, not on these.

Supplementary Results 4. A transcriptome-wide age predictor tracks proliferation (post hoc)

The same reading applies to age predictors built from these cells. A ridge model trained and cross-validated on all expressed genes in the 107 donors predicts chronological age with $r = 0.72$ (95% CI 0.61 to 0.80; mean absolute error 14 years), far outside a permuted-age null (20 permutations, $r = -0.31$ to 0.40; Supplementary Table 3). Its predictions track the proliferation score at $r = -0.74$, and proliferation accounts for 54% of the variance in predicted age (Supplementary Table 3), while the 20-gene proliferation score by itself predicts age at $r = 0.46$ (mean absolute error 18 years). A model built after removing the cohort covariates and proliferation

from every gene gives $r = -0.53$, but its predictions are nearly constant (57.5 to 59.0 years) and its error equals that of assigning every donor the cohort's mean age (21.6 against 21.5 years), so it carries no information about age. That arm cannot be read as isolating proliferation either: removing the cohort's technical covariates alone already lowers accuracy to $r = 0.24$, and repository membership cannot be separated from proliferation in this collection.

Supplementary Results 5. Reprogramming media and secretomes (post hoc class tests)

Interventions intended to rejuvenate fibroblasts almost all make them divide faster¹⁸, which makes them the sharpest test of the coupling. Seven reprogramming contrasts — naive and primed media, NHSM, RSeT and 5iLAF formulations¹⁹, OSK induction in fibroblasts from a 96-year-old donor and the matched arm in which wild-type OCT4 is replaced by the point mutant OCT4YR — and fourteen cell-free secretome preparations applied to recipient fibroblasts were examined on the same terms. One contrast previously counted as reprogramming was reclassified: in that series both the day-3 and the day-7 fibroblast samples are Sendai-OSKM-transduced cells still in fibroblast medium, the switch to reprogramming media having occurred on day 8, so their comparison measures four further days of culture.

Within each class the change in splicing-factor expression rises with the change in the cell cycle (reprogramming media $\rho = 0.89$, 95% CI 0.43 to 0.98, $P = 0.007$, $n = 7$ contrasts from two series; secretomes $\rho = 0.63$, 0.14 to 0.87, $P = 0.017$, $n = 14$ contrasts from eight series; Supplementary Fig. S4c). The within-class slope for reprogramming media is steeper than the slope outside the class (1.68, 95% CI 0.82 to 2.55, against 0.64) and depends on the naive-medium contrast (1.19, -0.64 to 3.02, without it). Predicted from a fit that excludes the class being predicted, 12 of the 21 interventions lie within 0.05 of the prediction and 7 lie more than 0.10 from it (Supplementary Fig. S4b): naive reprogramming medium (+0.34), NHSM (+0.12) and one serum-derived preparation (+0.14) above it, and OSK induction (-0.18), the OCT4YR arm (-0.14), a trophoblast stem cell secretome (-0.20) and a mesenchymal stromal cell secretome (-0.13) below it. In naive medium *PNISR*, *LSM14A*, *HNRNPD*, *SRSF7* and *HNRNPA1* rise by 0.91 to 1.44 in $\log_2$ (Supplementary Fig. S4c). Across the seven reprogramming contrasts *PNISR*, *SRSF2* and *TRA2B* rise in all seven (mean $\log_2$ 0.49 to 0.69), *HNRNPA1* and *HNRNPD* in five, *SRSF1* and *AKAP17A* barely move, and *IMP3* falls (mean -0.45).

Across all 21 intervention contrasts the median absolute residual is 0.045 against a median absolute change of 0.111 when each class is predicted from a fit that excludes it (0.038 from the in-sample fit of all 63 contrasts), so about three-fifths of a typical intervention effect is what its effect on the

cell cycle predicts. The share differs by class: about half for secretomes (0.041 against 0.082) and about one-ninth for reprogramming media (0.120 against 0.135), whose individual conditions depart from the line in both directions while their class mean does not (+0.02, 95% CI -0.14 to +0.18). With seven contrasts from two series, five of them sharing one reference arm, that class test could not detect a shift smaller than about 0.22, which is larger than the class's typical effect. The secretome class mean is -0.02 (-0.07 to +0.02, $n = 14$ contrasts from eight series; detectable shift about 0.07). For secretomes about half of a typical change in splicing-factor transcripts is what the cell-cycle response predicts; for reprogramming media most of it is not, in either direction. A rise in splicing-factor transcripts after an intervention therefore cannot be interpreted without a division-rate measurement alongside the transcriptome.

Supplementary Results 6. Which part of the expression machinery lies closest to division (post hoc)

Splicing is not the only part of the machinery of gene expression that follows the cell cycle, so the splicing set was compared with three Reactome sets scored the same way: Translation, Metabolism of RNA and rRNA processing. The 177-gene set was selected for declining in three datasets, one of which compares late with early passages in the very resource where division rate falls with passage. The comparison was therefore run with two unselected definitions as well — Reactome mRNA Splicing (215 genes) and Processing of Capped Intron-Containing Pre-mRNA (296 genes) — and with the pathway genes lying outside the 177 as a lower bound. No comparison shares a gene: shared genes were removed from the comparator for the selected set and from both sides for the other definitions (removing them from both sides for the selected set as well changes no partial correlation by more than 0.01), except that Metabolism of RNA contains both splicing pathways, so there the splicing genes were removed from the comparator only.

Against the counted division rate (328 libraries) every splicing definition follows the rate more closely than the comparators do (selected set $\rho = 0.68$, mRNA Splicing 0.63, pre-mRNA processing 0.65; comparators 0.43 to 0.52). The selected set reaches 0.71 with the score of Fig. 2a; here every set is scored in the expression matrix of the programme audit so that all sets share one filter, and 176 of the 177 genes pass it. Holding a comparator constant leaves the splicing definition with most of its coupling (partial $\rho = 0.56$ to 0.59 for the selected set, 0.43 to 0.51 for mRNA Splicing, 0.52 to 0.55 for pre-mRNA processing), while holding the splicing definition constant leaves the comparators with none (-0.22 to +0.01; Supplementary Fig. S5a). With the unselected definitions the asymmetry is 66 to 91% of its size with the selected set. It is carried by the selected genes: for the 119

pre-mRNA processing genes outside the 177 it reverses (splicing partial -0.07 to $+0.15$, comparators 0.30 to 0.50).

In the 107 donors the same holds for the age effect. Selection inflated only the unadjusted effect (-0.133 per decade for the selected set against -0.094 for mRNA Splicing); the proliferation-adjusted residue is the same (-0.040 and -0.042), and with translation also in the model it is -0.022 ($P = 0.044$) and -0.023 ($P = 0.015$), whereas the decline of translation and of Metabolism of RNA disappears once either splicing score is in the model (-0.001 to $+0.009$, $P = 0.17$ to 0.94 ; Supplementary Fig. S5c).

GTEx cultures do not reproduce the asymmetry. There proliferation is the 20-marker transcriptional score, not a counted rate, and against it translation lies closer than splicing does: holding the other constant, translation keeps 0.51 to 0.59 and the splicing definitions 0.28 to 0.41 ($n = 511$ cultures; Supplementary Fig. S5b). Against Metabolism of RNA and against rRNA processing the selected set keeps an asymmetry (0.60 against -0.29 , and 0.53 against 0.26) but the unselected definitions do not (differences within their bootstrap intervals). The asymmetry therefore appears where division was counted and in the donor cohort, and not against a transcriptional proliferation score in an independent collection. These scores are strongly correlated with one another ($r = 0.68$ to 0.89 in the counted libraries), so what the comparison can establish is which of them lies closest to division rate, not that they are independently regulated programmes (Supplementary Table 5).

Supplementary Results 7. Seven senescence panels (post hoc)

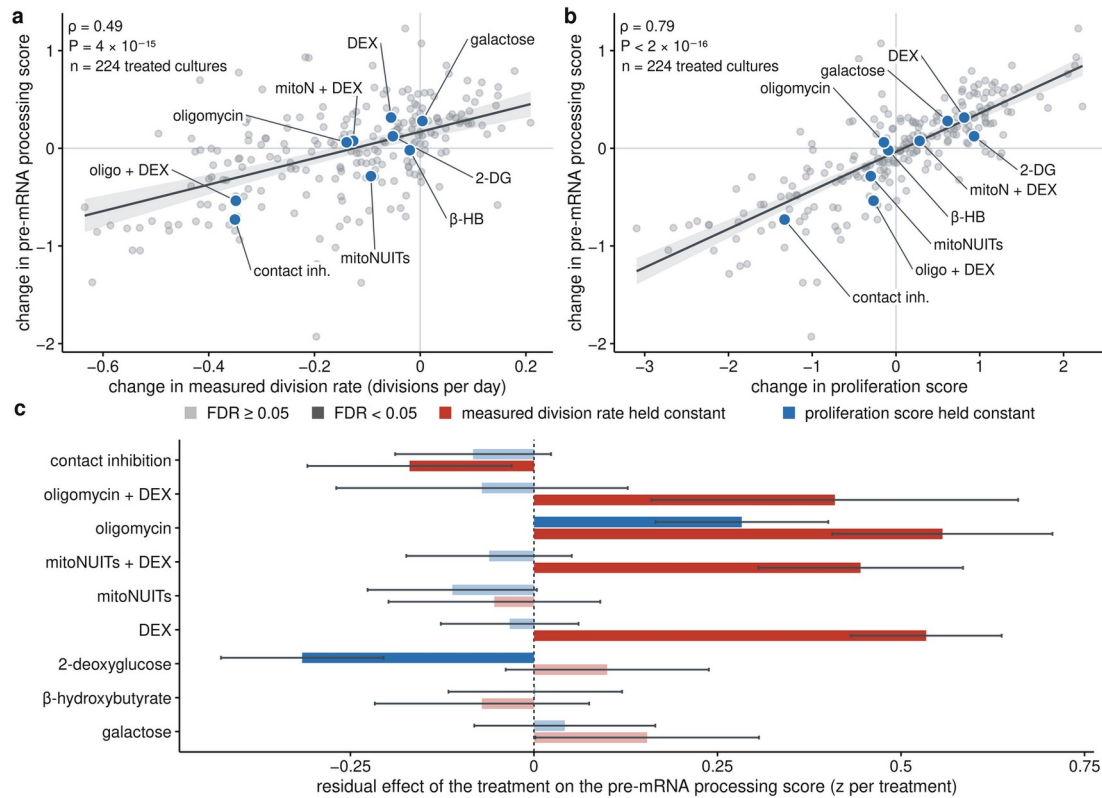
Because the choice of panel decides the answer, five further panels were scored the same way as the Reactome set: SenMayo²², the up and down halves of the fibroblast senescence signature of Fridman and Tainsky³⁶, and the CellAge genes annotated as inducing or as inhibiting senescence³⁷, with the Reactome mitotic cell cycle as the positive control. Across the seven, panels with a larger share of cell-cycle genes tend to correlate more strongly with the measured division rate ($\rho = 0.71$, exact two-sided $P = 0.088$, $n = 7$ panels; Supplementary Fig. S6a); with seven panels the trend does not reach significance. Reactome Cellular Senescence (55% cell-cycle genes) reaches $\rho = +0.67$ and falls to $+0.38$ when those genes are removed; SenMayo (1%) -0.10 ; Fridman up (15%) -0.07 ; Fridman down (27%, eleven genes measured) $+0.38$, falling to $+0.01$; the CellAge inducers (10%) $+0.33$. The CellAge genes annotated as inhibiting senescence do not fit the composition account: 21% of them are cell-cycle genes, yet the panel reaches $+0.69$ and keeps $+0.71$ after those genes are removed, as might be expected of genes whose activity lets cells keep dividing.

In the 107 donors, five of the seven panels have an age association before adjustment ($FDR < 0.05$): Reactome Cellular Senescence (-0.026 per decade), the CellAge inhibitors (-0.041), Fridman down (-0.034) and the cell-cycle control (-0.088) decline, and Fridman up rises ($+0.043$). SenMayo and the CellAge inducers have none ($FDR = 0.72$). None of the five keeps its association once proliferation is included; they lose 66 to 100% of the estimate (Supplementary Fig. S6b, Supplementary Table 7).

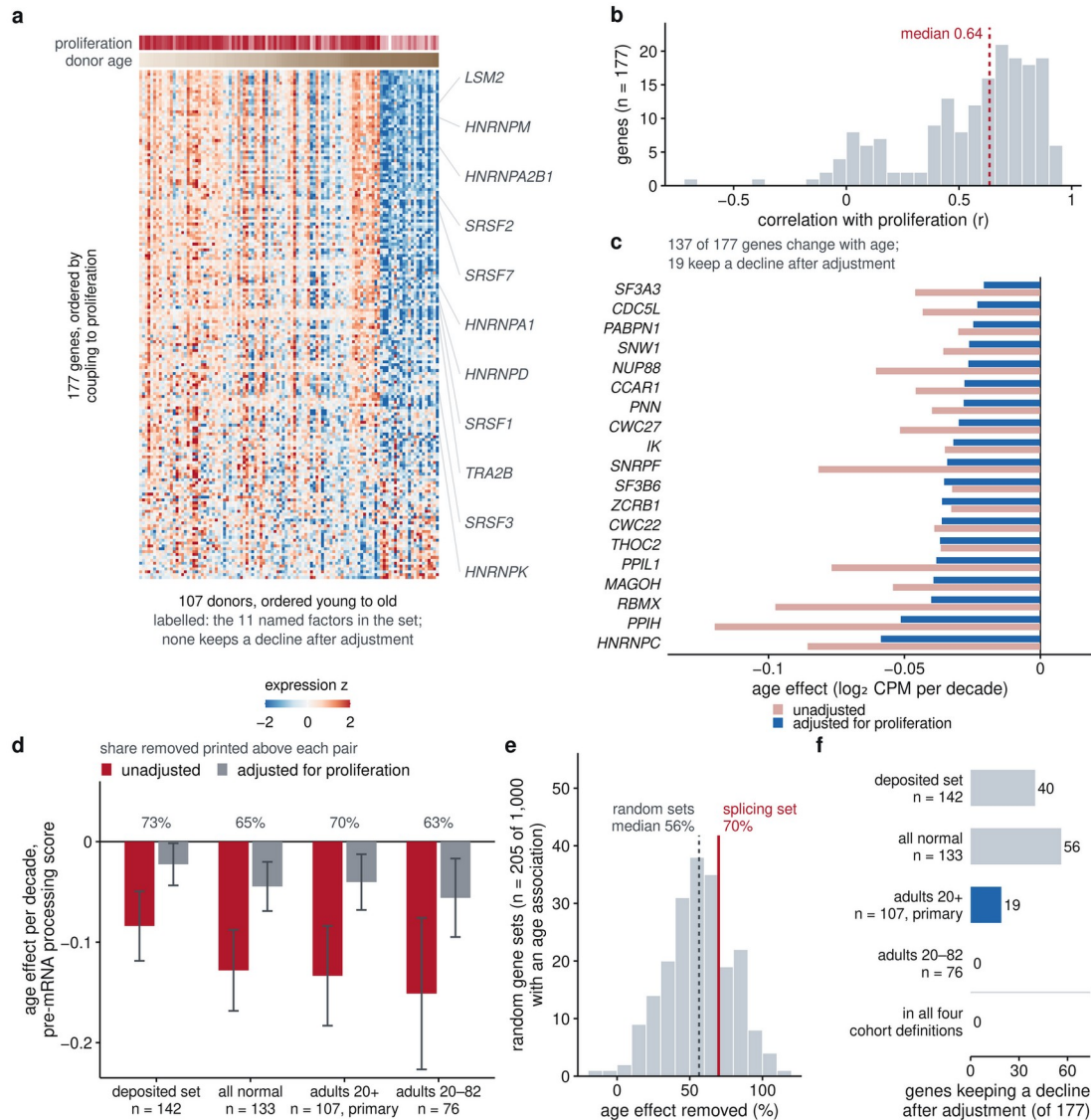
Supplementary Results 8. A donor's cultured fibroblasts do not track that donor's skin

Programme-level state is not shared between a donor's cultured fibroblasts and that donor's skin. Among 388 donors who contributed both cultured fibroblasts and skin, no programme score in the cultured cells was related to the same programme in that donor's skin ($\rho = -0.10$ to 0.13 , none with $FDR < 0.05$; Supplementary Fig. S8c); this concordance was exploratory analysis S3 of the preregistration, for which no success criterion was set (Supplementary Note 1). That is not a failure of the measurement: in the same donors and the same matrices, genes whose expression is strongly genotype-dependent are recovered across the two sample types (*GSTM1* $\rho = 0.75$, *ERAP2* $\rho = 0.75$, both $P < 10^{-70}$; seven sex-chromosome genes 0.67 to 0.74); this positive control was added post hoc. Programme scores measured in cultured fibroblasts therefore carry no detectable information about the same programmes in the donor's bulk skin ($|\rho| \leq 0.13$, $n = 388$ donors); because bulk skin is mostly not fibroblasts, this comparison cannot separate the effect of culture from that of cell type.

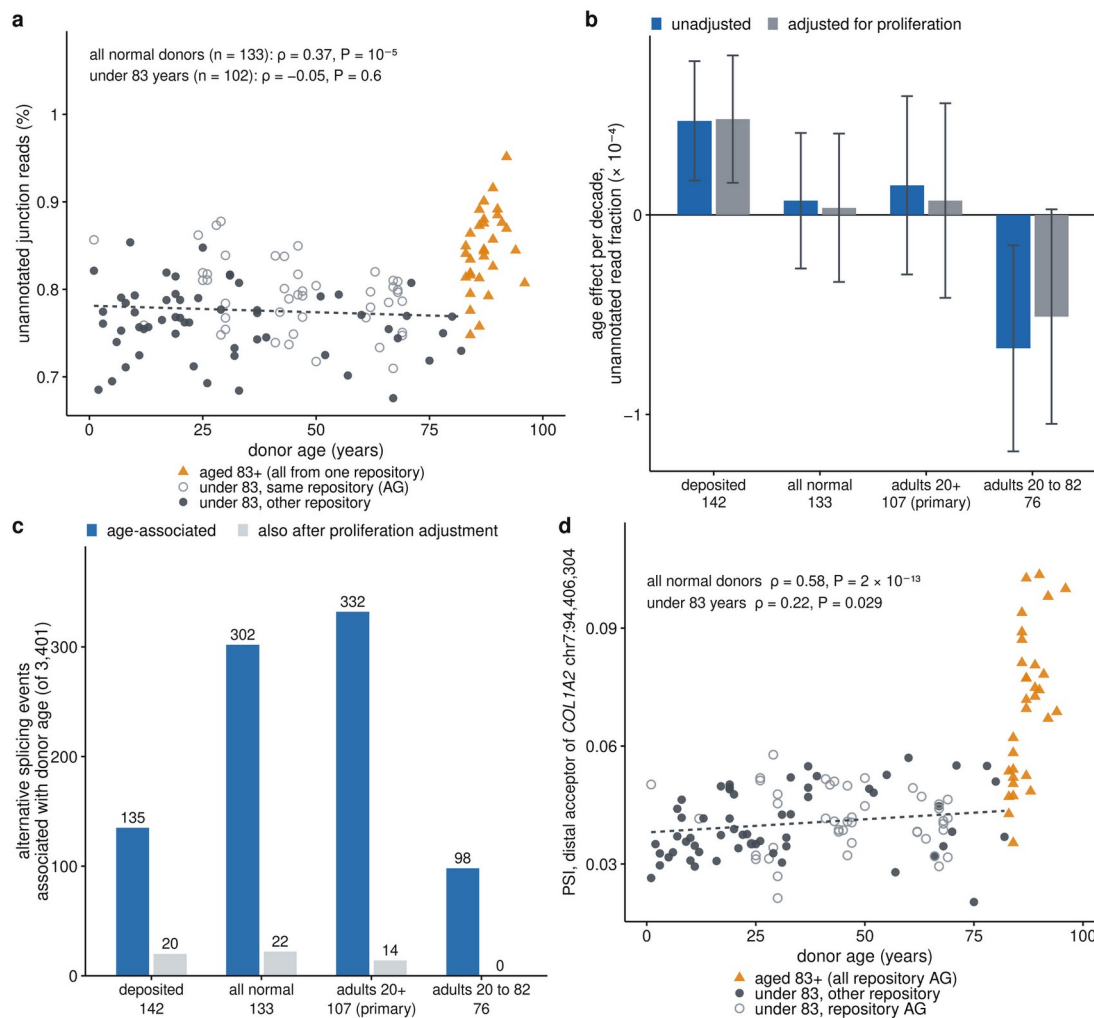
Supplementary Figures



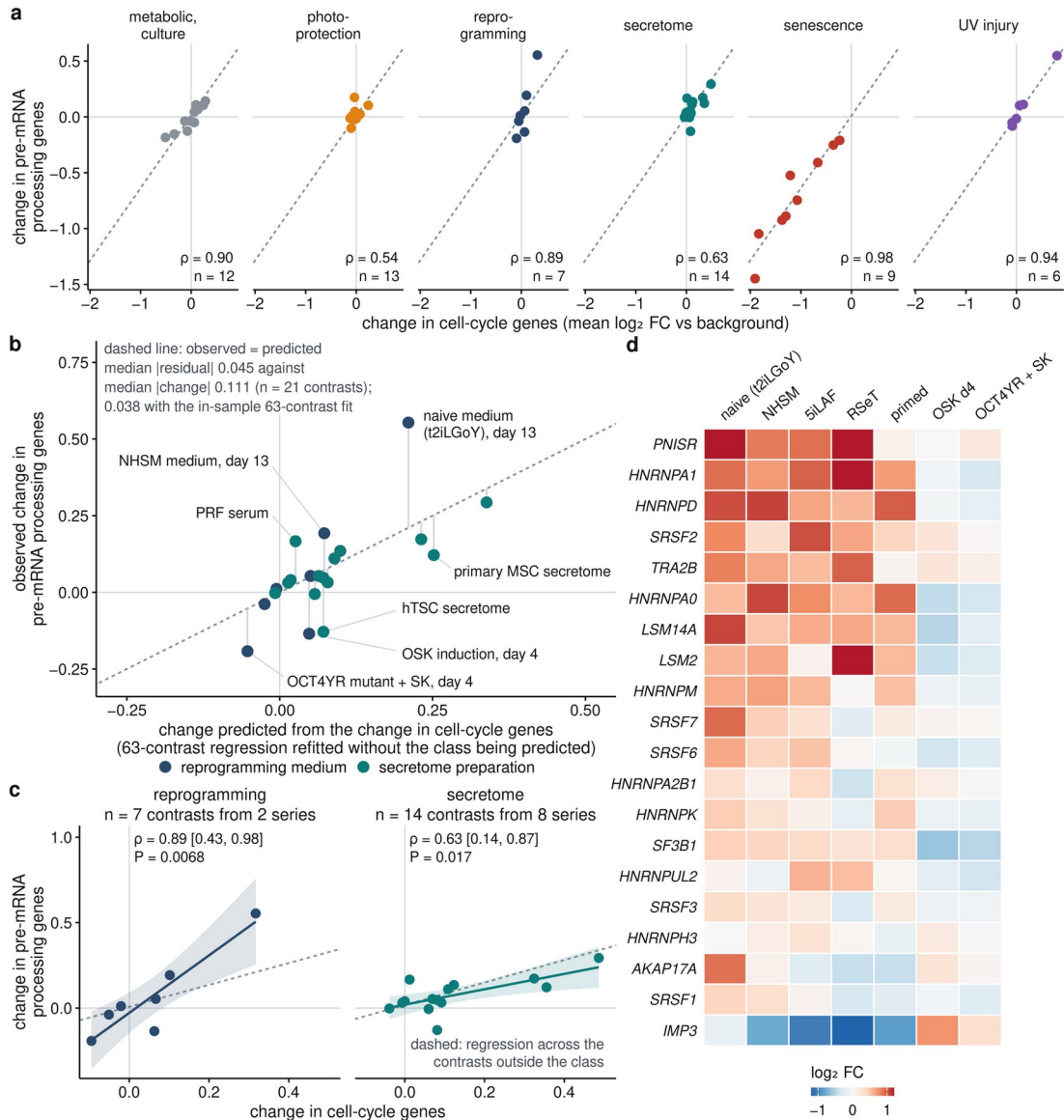
Supplementary Fig. S1. The coupling under nine perturbations of the same cell lines (post hoc). (a) For each treated culture, the difference from the control culture of the same cell line and oxygen tension nearest in days grown: change in measured division rate against change in the 177-gene splicing score. Grey, individual cultures; blue, the median of each treatment arm, with the arms discussed in Supplementary Results 1 labelled. (b) The same against the change in the transcriptional proliferation score. (c) Residual effect of each treatment on the splicing score, with cell line, oxygen tension and days grown in the model and either the measured division rate or the proliferation score held constant; bars with 95% confidence intervals, bars at reduced opacity FDR ≥ 0.05 , treatments ordered by how much they lower the division rate. Values, and the oxygen contrast, are in Supplementary Table 2.



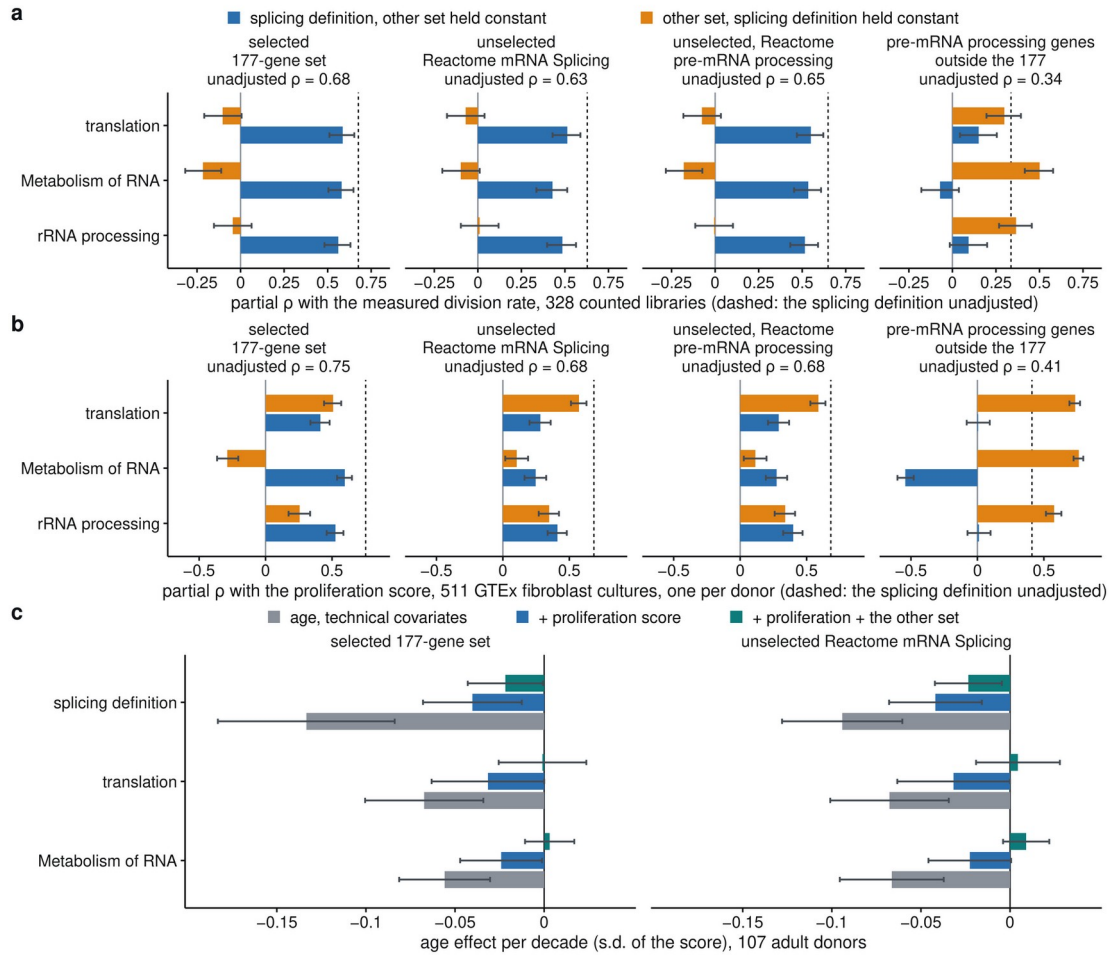
Supplementary Fig. S2. Which genes keep an age association. (a) Expression of the 177 genes across the 107 donors ordered young to old, genes ordered by their own correlation with proliferation; bars above show donor age and the proliferation score; the eleven of the twenty named factors that belong to the set are labelled, none of which keeps an age association after adjustment. (b) Distribution of each gene's correlation with the proliferation score. (c) The 19 genes keeping a decline, with the age effect in log₂ counts per million per decade before (pink) and after (blue) adjustment. (d) Age effect per decade for four cohort definitions. (e) Distribution of random gene sets with an age association. (f) Number of genes keeping a decline after adjustment for different cohort definitions.



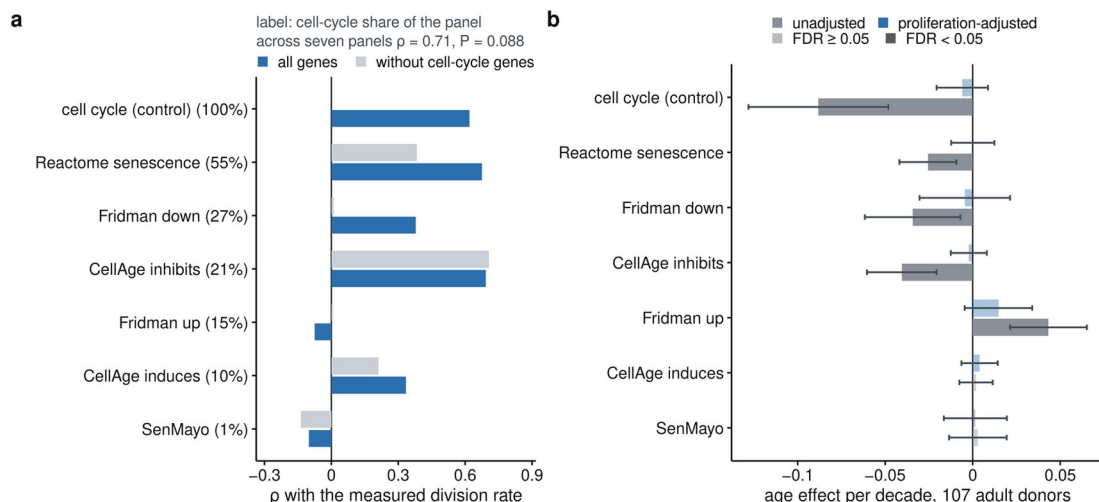
Supplementary Fig. S3. The splicing outcome index and event-level splicing, neither of which is interpreted. (a) The share of junction reads on unannotated junctions against donor age in all 133 normal donors, with the 31 donors aged 83 and over — every one from the same cell repository — drawn as triangles and the younger donors of that repository as open symbols. (b) Age effect per decade on the unannotated-read fraction, before and after adjustment for proliferation, under each of the four cohort definitions, as bars with 95% confidence intervals; the deposited set uses sequencing depth as its only covariate. Only the deposited 142-sample set gives a positive age effect, and among adults aged 20 to 82 the estimate runs the other way. (c) Age-associated event counts under each cohort definition, before and after adjustment for proliferation. (d) The *COL1A2* event with the strongest age association in the deposited set against donor age, with the 83-and-over stratum and the cell repository marked; the distal junction that carries that signal is not annotated in GENCODE v41.



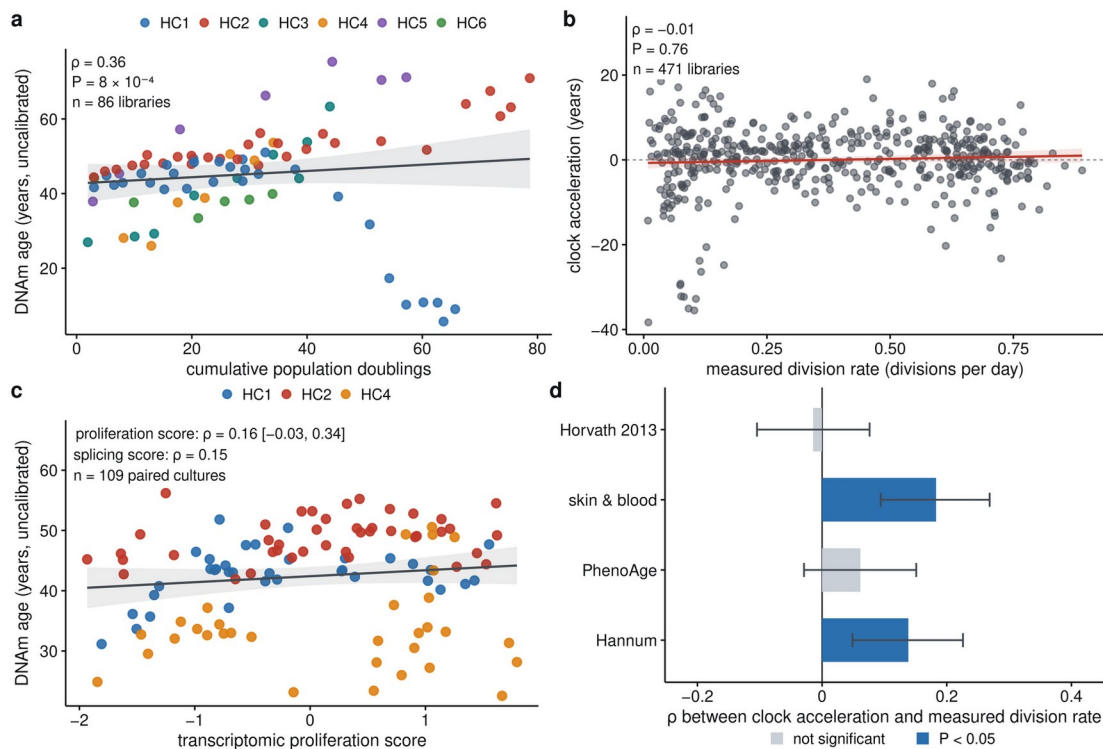
Supplementary Fig. S4. The coupling within each class of perturbation, and the post hoc class tests of reprogramming media and secretomes. (a) The relationship of Fig. 6a within each of the six classes of perturbation with at least five contrasts, on common axes; dashed line, the fit across all 63 contrasts. (b) Observed change in splicing-factor expression against the change predicted from the cell-cycle response, for seven reprogramming contrasts and fourteen secretome preparations; each class is predicted from the 63-contrast regression refitted without that class. (c) The relationship within each intervention class, with the number of contrasts and of series; dashed line, the regression outside the class. (d) The twenty named splicing factors across the seven reprogramming contrasts, ordered by mean response; grey cells are genes absent from the gene table of that contrast.



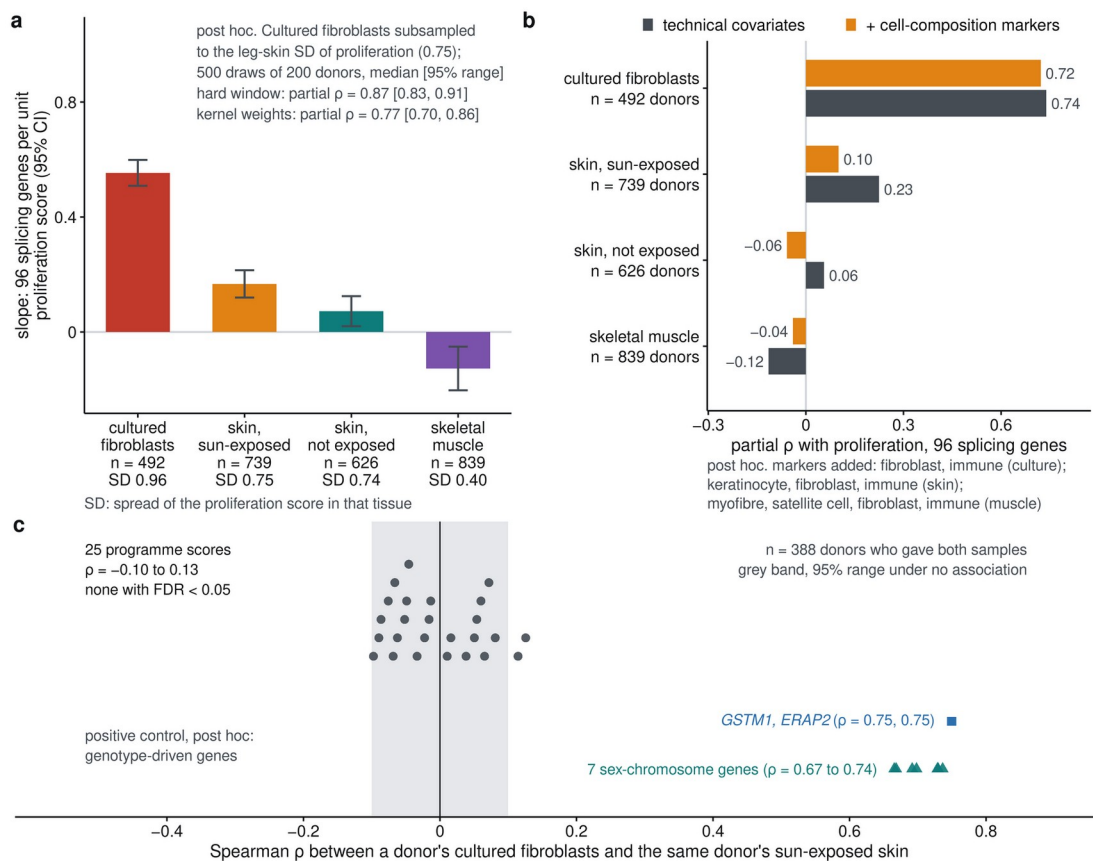
Supplementary Fig. S5. Which part of the expression machinery lies closest to division (post hoc). No comparison shares a gene. **(a)** Partial Spearman correlation with the measured division rate across the 328 counted libraries, for the selected 177-gene set, two unselected Reactome definitions and the pathway genes outside the 177, as bars with 95% confidence intervals: the splicing definition with each comparator held constant (blue) and each comparator with the splicing definition held constant (orange); dashed line, the definition's unadjusted correlation in this matrix (0.68 for the 177-gene set, against 0.71 for the score of Fig. 2a; Methods). **(b)** The same comparison against the 20-marker proliferation score in 511 GTEx cultured fibroblasts, where the asymmetry is not reproduced. **(c)** Age effect per decade in the 107 adult donors for the selected set and for unselected Reactome mRNA Splicing, with the cohort's technical covariates, then with the proliferation score, then with the proliferation score and the other set.



Supplementary Fig. S6. Seven senescence panels (post hoc). (a) For seven panels scored identically in the 328 counted libraries, the share of measured genes that are also cell-cycle genes printed in the label, and the panel score's correlation with the measured division rate as bars: dark, all genes; light, after every cell-cycle gene is removed; the cell-cycle control has nothing left to score after removal. (b) Age effect per decade for the same panels in the 107 adult donors, before and after adjustment for proliferation, as bars with 95% confidence intervals, bars at reduced opacity $\text{FDR} \geq 0.05$; panels ordered by cell-cycle content.



Supplementary Fig. S7. Preregistered tests of methylation clocks in the same cultures. (a) DNAm age from Horvath's multi-tissue clock against the cumulative population doublings of untreated healthy lines at 21% oxygen; absolute values are uncalibrated approximations and only ranks are interpreted. (b) Clock acceleration — DNAm age with days in culture regressed out within each cell line — against the measured division rate across 471 libraries from nine cell lines and nineteen treatments. (c) DNAm age against the transcriptomic proliferation score in the 109 cultures carrying both measurements, with the splicing score's correlation given alongside; colour, cell line. (d) Correlation between clock acceleration and measured division rate for four published clocks, as bars with 95% confidence intervals from Fisher's z ; blue, $P < 0.05$. Supporting comparison, not among the preregistered tests: M2 was fixed for clock acceleration without naming a clock; Horvath's clock, the one the preregistration named unconditionally, carries the recorded verdict, the skin-and-blood clock and PhenoAge were named conditionally on their coefficient tables and each also falls below the preregistered 0.3, and Hannum's clock was added after the preregistration was locked (Supplementary Note 2). M1 (a) met its criterion; M2 (b), M3 (c) and M4 (not drawn; Supplementary Table 8) did not.



Supplementary Fig. S8. Alternatives to the tissue result, and donor-level concordance. (a) Post hoc: the regression slope of the splicing score on the proliferation score in each tissue as bars with 95% confidence intervals, the spread (SD) of the proliferation score in each tissue given below, and the partial correlation obtained when cultured fibroblasts are subsampled to the proliferation spread of skin by two schemes (a hard window and kernel weights). (b) Post hoc: the splicing-proliferation partial correlation in each tissue before and after adjustment for cell-composition marker scores (bars, values printed; the markers added in each tissue are listed under the panel). (c) Spearman correlation, across the 388 donors who gave both, between each programme's score in a donor's cultured fibroblasts and in the same donor's skin (preregistered exploratory analysis, no success criterion; Supplementary Note 1), with two genes whose expression is strongly genotype-dependent and seven sex-chromosome genes as a positive control added post hoc.

Supplementary Tables

Supplementary Table 1. Cohort sensitivity in the donor series

Every donor-level result under four definitions of the GSE113957 cohort. Age effects are per decade of donor age; "lost" is the proportion of the unadjusted effect removed by adjustment for proliferation.

cohort	n	rho age- prol	effect	P	adj effect	adj P	% lost
adults 20+ (primary)	107	-0.46	-0.133	5.9e-07	-0.04	0.0049	70
all normal (133)	133	-0.45	-0.128	4.6e-09	-0.045	0.00044	65
adults 20-82	76	-0.25	-0.151	0.00015	-0.056	0.0057	63
deposited (142)	142	-0.4	-0.084	3.9e-06	-0.023	0.035	73

Gene-level counts and the outcome index in the same four cohorts.

cohort	unann. effect	unann. P	genes, age	genes kept	PSI events
adults 20+ (primary)	1.5e-05	0.51	137	19	332
all normal (133)	7.1e-06	0.68	152	56	302
adults 20-82	-6.7e-05	0.012	107	0	98
deposited (142)	4.7e-05	0.0022	129	40	135

cohort	unann. effect	unann. P	genes, age	genes kept	PSI events
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Supplementary Table 2. Treatment perturbation in the counted resource

The splicing score is the 177-gene score of Fig. 2. All of this table is post hoc.

(a) Treatment arms. Libraries and cell lines per arm. Arms with fewer than five libraries or present in only one cell line were to be left out of the models; no arm fell below these limits.

treatment	libraries	cell lines	modelled
Control	104	7	TRUE
DEX	54	7	TRUE
Contact_Inhibition	41	3	TRUE
mitoNUTs+DEX	24	3	TRUE
2-Deoxyglucose	24	2	TRUE
mitoNUTs	20	3	TRUE
Oligomycin	19	3	TRUE
betahydroxybutyrate	19	2	TRUE
Galactose	17	2	TRUE
Oligomycin+DEX	6	3	TRUE

(b) Residual effect of each treatment, with cell line, oxygen tension and days grown in every model and Benjamini-Hochberg correction within each model. A treatment is counted as keeping a residual effect when its coefficient is significant at BH-FDR < 0.05 within the model, and as accounted for by the adjustment named in the first column otherwise. Counts in the text and filled symbols in Supplementary Fig. S1c use BH-FDR < 0.05.

response and adjustment	treatment	effect	95% CI lower	95% CI upper	P	FDR
splice, rate not in model	2- Deoxyglucose	-0.077	-0.226	0.073	0.31	0.31
splice, rate not in model	betahydroxybutyrate	-0.113	-0.276	0.051	0.18	0.23

response and adjustment	treatment	effect	95% CI lower	95% CI upper	P	FDR
splice, rate not in model	Contact_In hibition	−0.530	−0.659	−0.400	2.5e-14	2.3e-13
splice, rate not in model	DEX	0.401	0.291	0.512	8.8e-12	4e-11
splice, rate not in model	Galactose	0.144	−0.027	0.316	0.099	0.15
splice, rate not in model	mitoNUITs	−0.144	−0.305	0.016	0.079	0.14
splice, rate not in model	mitoNUITs +DEX	0.248	0.098	0.397	0.0013	0.0029
splice, rate not in model	Oligomycin	0.384	0.221	0.547	5.9e-06	1.8e-05
splice, rate not in model	Oligomycin +DEX	0.164	−0.110	0.438	0.24	0.27
splice, measured rate held constant	2- Deoxygluco se	0.100	−0.038	0.238	0.16	0.2
splice, measured rate held constant	betahydrox ybutyrate	−0.071	−0.217	0.075	0.34	0.39
splice, measured rate held constant	Contact_In hibition	−0.170	−0.309	−0.030	0.017	0.031
splice, measured rate held constant	DEX	0.534	0.432	0.637	3e-21	2.7e-20
splice, measured rate held	Galactose	0.154	0.002	0.306	0.048	0.072

response and adjustment	treatment	effect	95% CI lower	95% CI upper	P	FDR
constant						
splice, measured rate held constant	mitoNUITs	−0.054	−0.198	0.090	0.46	0.46
splice, measured rate held constant	mitoNUITs +DEX	0.445	0.306	0.584	1.3e-09	3.9e-09
splice, measured rate held constant	Oligomycin	0.556	0.406	0.706	2.9e-12	1.3e-11
splice, measured rate held constant	Oligomycin +DEX	0.410	0.160	0.659	0.0014	0.0032
proliferatio n score, measured rate held constant	2- Deoxygluco se	0.876	0.646	1.106	8.9e-13	2.7e-12
proliferatio n score, measured rate held constant	betahydrox ybutyrate	−0.203	−0.445	0.040	0.1	0.12
proliferatio n score, measured rate held constant	Contact_In hibition	−0.454	−0.685	−0.223	0.00015	0.00022
proliferatio n score, measured rate held constant	DEX	1.270	1.099	1.441	5.5e-37	4.9e-36
proliferatio n score, measured	Galactose	0.262	0.009	0.516	0.043	0.055

response and adjustment	treatment	effect	95% CI lower	95% CI upper	P	FDR
rate held constant						
proliferatio n score, measured rate held constant	mitoNUITs	0.076	-0.164	0.316	0.54	0.54
proliferatio n score, measured rate held constant	mitoNUITs +DEX	1.079	0.847	1.311	9.6e-18	4.3e-17
proliferatio n score, measured rate held constant	Oligomycin	0.537	0.288	0.787	3.2e-05	5.7e-05
proliferatio n score, measured rate held constant	Oligomycin +DEX	0.985	0.569	1.400	5e-06	1.1e-05
splice, proliferatio n score held constant	2- Deoxygluco se	-0.316	-0.426	-0.205	5e-08	4.5e-07
splice, proliferatio n score held constant	betahydrox ybutyrate	0.002	-0.116	0.120	0.98	0.98
splice, proliferatio n score held constant	Contact_In hibition	-0.083	-0.189	0.023	0.13	0.29
splice, proliferatio n score	DEX	-0.033	-0.127	0.061	0.49	0.57

response and adjustment	treatment	effect	95% CI lower	95% CI upper	P	FDR
held constant						
splice, proliferatio n score	Galactose	0.042	-0.081	0.165	0.5	0.57
held constant						
splice, proliferatio n score	mitoNUITs	-0.111	-0.227	0.004	0.059	0.18
held constant						
splice, proliferatio n score	mitoNUITs +DEX	-0.061	-0.174	0.052	0.29	0.52
held constant						
splice, proliferatio n score	Oligomycin	0.283	0.166	0.401	3.6e-06	1.6e-05
held constant						
splice, proliferatio n score	Oligomycin +DEX	-0.071	-0.269	0.128	0.49	0.57
held constant						

(c) Matched treated-minus-control differences, median per arm, against control cultures of the same cell line and oxygen tension nearest in days grown.

treatment	cultu res	delta division rate	delta splicing	delta prolifer ation	medi an delta days
Contact_Inhibition	41	-0.350	-0.73	-1.33	4.9
Oligomycin+DEX	6	-0.349	-0.54	-0.27	0.0
Oligomycin	19	-0.139	0.06	-0.14	0.0
mitoNUITs+DEX	24	-0.127	0.08	0.28	0.0

treatment	cultures	delta division rate	delta splicing	delta proliferation	median delta days
mitoNUTs	20	-0.093	-0.29	-0.30	0.0
DEX	54	-0.055	0.31	0.81	0.0
2-Deoxyglucose	24	-0.052	0.12	0.93	0.0
betahydroxybutyrate	19	-0.020	-0.02	-0.09	0.0
Galactose	17	0.004	0.28	0.62	0.0

(d) Oxygen tension. Cultures at 3% oxygen against cultures of the same cell line at 21% oxygen: medians of the differences (3% minus 21%), with the number of cultures in which the 3% value is the lower one. The unmatched rows reproduce a pooled comparison and are given so that its composition can be seen.

contrast	line	n at 3%	n at 21%	delta rate	delta splicing	delta proliferation	median delta day	rate lower	splicing lower
control, 3% minus 21% oxygen, same line, nearest days grown	all	15	14	-0.099	-0.51	-0.26	2.3	12	12
control, 3% minus 21% oxygen, same line, nearest days grown	HC 4	5	4	-0.019	-0.51	0.04	0.4	3	4
control, 3% minus 21% oxygen, same line, nearest days grown	HC 1	5	5	-0.180	-0.21	-0.05	2.3	5	3
control, 3% minus 21% oxygen, same line, nearest days grown	HC 2	5	5	-0.106	-0.59	-0.39	2.5	4	5

contrast	line	n at 3%	n at 21 %	delta rate	delta splici ng	delta prolif eratio n	me dia n delt a day s	rate low er	spli cin g low er
control, 3% minus 21% oxygen, matches within 7 days grown only	all	13	13	-0.099	-0.51	-0.26	2.3	11	11
contact inhibition, 3% minus 21% oxygen, same line, nearest days grown	all	20	20	-0.002	-0.39	0.75	0.0	15	19
control, unmatched: median of 3% minus median of 21%, same lines, all days	all	15	63	0.052	-0.22	0.77			
control, unmatched, 21% cultures within 7 days of the 3% time span	all	15	38	-0.047	-0.52	0.20			
older sentence (run_division _rate.R): every 3% library minus every 21% library of the same lines, treatments pooled	all	35	229	-0.017	-0.40	-0.38			

The oxygen term in models of control cultures, with cell line and days grown as covariates.

libraries	response	held constant	n	n at 3%	effect of 3% oxygen	95% CI lower	95% CI upper	P
control cultures, all seven lines	measured division rate	none	104	15	-0.078	-0.133	-0.024	0.0061
control cultures, all seven lines	proliferation score	none	104	15	-0.055	-0.413	0.302	0.76
control cultures, all seven lines	splicing score	none	104	15	-0.361	-0.561	-0.161	0.00064
control cultures, all seven lines	splicing score	measured division rate	104	15	-0.213	-0.392	-0.034	0.022
control cultures, all seven lines	splicing score	proliferation score	104	15	-0.335	-0.442	-0.228	2e-08

libraries	response	held constant	n	n at 3%	effect of 3% oxygen	95% CI lower	95% CI upper	P
lines								
control cultures, all seven lines	proliferation score	measured division rate	104	15	0.177	-0.156	0.510	0.3
control cultures, the three lines grown at both oxygen tensions	measured division rate	none	78	15	-0.075	-0.129	-0.021	0.0084
control cultures, the three lines grown at both oxygen tensions	proliferation score	none	78	15	-0.046	-0.408	0.317	0.8
control	splicing score	none	78	15	-0.353	-0.551	-0.155	0.0008

libraries	response	held constant	n	n at 3%	effect of 3% oxygen	95% CI lower	95% CI upper	P
cultures, the three lines grown at both oxygen tensions								
control cultures, the three lines grown at both oxygen tensions	splicing score	measured division rate	78	15	−0.215	−0.395	−0.035	0.022
control cultures, the three lines grown at both oxygen tensions	splicing score	proliferation score	78	15	−0.331	−0.428	−0.234	4e-09

libraries	response	held constant	n	n at 3%	effect of 3% oxygen	95% CI lower	95% CI upper	P
ons								
control cultures, the three lines grown at both oxygen tensions	proliferation score	measured division rate	78	15	0.163	-0.185	0.511	0.36
all modelled libraries (the primary treatment model)	measured division rate	none	328	35	-0.011	-0.059	0.038	0.67
all modelled libraries (the primary treatment model)	proliferation score	none	328	35	0.405	0.179	0.632	0.00052
all	splicin	none	328	35	-0.317	-0.453	-0.182	6.3e-06

libraries	response	held constant	n	n at 3%	effect of 3% oxygen	95% CI lower	95% CI upper	P
modelled libraries (the primary treatment model)	g score							
all modelled libraries (the primary treatment model)	splicing score	measured division rate	328	35	−0.304	−0.424	−0.183	1.3e-06
all modelled libraries (the primary treatment model)	splicing score	proliferation score	328	35	−0.486	−0.585	−0.387	2.3e-19
all modelled libraries (the primary treatment model)	proliferation score	measured division rate	328	35	0.429	0.228	0.629	3.6e-05

libraries	response	held constant	n	n at 3%	effect of 3% oxygen	95% CI lower	95% CI upper	P
ary								
treat								
ment								
mod								
el)								

Supplementary Table 3. Sensitivity analyses in the donor cohort

(a) Technical quality covariates. Uniquely mapped percentage, multi-mapping percentage, intronic signal and exon assignment rate added to the cohort model.

metric	covariates	beta	p	beta adj	p adj	pct lost
splicing-factor expression	cohort	-0.133	5.9e-07	-0.0402	0.0049	70
splicing-factor expression	cohort + QC	-0.11	7.4e-06	-0.0386	0.0029	65
unannotated junction reads	cohort	1.48e-05	0.51	7.19e-06	0.77	-
unannotated junction reads	cohort + QC	1.08e-06	0.96	8.87e-06	0.7	-
entropy of junction usage	cohort	-4.1e-05	0.68	0.000156	0.1	-
entropy of junction usage	cohort + QC	4.03e-05	0.68	0.000191	0.041	-

(b) Core spliceosome versus the rest of the set. The 127 genes belonging to Reactome mRNA splicing, and the 50 genes of capping, 3'-end processing and export.

metric	covariates	beta	p	beta adj	p adj	pct lost
splicing-factor	cohort	-0.133	5.9e-07	-0.040	0.0049	70

metric	covariates	beta	p	beta adj	p adj	pct lost
expression						
core spliceosome (127 genes)	cohort	-0.132	4.8e-07	-0.044	0.0043	66
capping, 3'-end, export (50 genes)	cohort	-0.136	2.4e-06	-0.030	0.022	78

The same three scores against the measured division rate in the 328 counted libraries (Spearman correlation with its 95% confidence interval), all scored in the expression matrix of the programme audit.

gene set	genes measured	libraries	rho with division rate	95% CI lower	95% CI upper	P
whole set (177 genes)	176	328	0.68	0.61	0.73	3.1e-45
core spliceosome (127 genes)	126	328	0.66	0.60	0.72	9.9e-43
capping, 3'-end, export (50 genes)	50	328	0.70	0.64	0.75	6.4e-49

(c) Base coverage versus read equivalents. Gene sums divided by read length (51 or 75 nucleotides in this series).

metric	covariates	beta	p	beta adj	p adj	pct lost	scale used
splicing-factor expression	cohort	-0.133	5.9e-07	-0.040	0.0049	70	base coverage
splicing-factor expression	cohort	-0.134	6.7e-07	-0.039	0.0055	71	read equivalents

(d) Transcriptome-wide age predictor (post hoc). Ten-fold cross-validation repeated five times in the 107 donors; the permuted-age null is twenty repetitions.

model	r	95% CI lower	95% CI upper	MAE	r with proliferation score
all genes	0.72	0.61	0.80	14.2	−0.74
all genes, cohort covariates removed	0.24	0.06	0.42	21.2	−0.75
proliferation score alone	0.46	0.29	0.59	18.5	−1.00
all genes, proliferation removed	−0.53	−0.66	−0.38	21.6	0.25
permuted age (one draw)	−0.28	−0.45	−0.10	21.9	0.44

(e) Negative control for the proliferation adjustment. The unadjusted and the proliferation-adjusted cohort models applied to 1,000 random gene sets matched to the splicing set for expression level and containing no splicing or cell-cycle genes. The share removed is given for the random sets that have an age association of their own, because it is undefined without one; effects are in the units of the splicing score.

quantity	value
random gene sets drawn	1000
sets with an age association of their own (unadjusted $P < 0.05$)	205
of those, increasing with age / decreasing with age	184 / 21
share of their age effect removed by adjustment, %: median	56
the same, interquartile range	43 to 71
the same, 2.5th to 97.5th percentile	13 to 98
of those, sets losing at least as much as the splicing set (69.8%)	55 of 205 (27%)
of those, sets keeping an age effect after adjustment (adjusted $P < 0.05$)	10 of 205
unadjusted age effect per decade, range over all sets	−0.018 to +0.030
smallest unadjusted P over all sets	7.2e-05
smallest adjusted P over all sets	0.01
splicing set, unadjusted age effect per decade (P)	−0.133 (5.9e-07)
splicing set, adjusted age effect per decade (P)	−0.040 (0.0049)
random sets with an unadjusted effect at least	0

quantity	value
as large as the splicing set's	
random sets with an unadjusted P at least as small	0
random sets with an adjusted P at least as small	0

(f) Which technical covariate the residual depends on. The proliferation-adjusted age effect of the 177-gene score in the 107 donors, per decade with its 95% confidence interval, under subsets of the cohort's technical covariates; the share removed is relative to the unadjusted model that carries the same covariates. The two technical terms overlap: 13 of the 107 libraries were sequenced on the HiSeq 2500, all of them from the AG repository.

covariates	unadjusted	P	adjusted	95% CI (adjusted)	P (adj)	% removed
depth	-0.099	6.6e-05	-0.022	-0.049 to 0.005	0.11	78
depth + repository	-0.116	3.3e-06	-0.038	-0.063 to -0.012	0.0038	68
depth + instrument	-0.101	8.7e-05	-0.015	-0.043 to 0.013	0.29	85
depth + sex	-0.105	2.6e-05	-0.025	-0.053 to 0.002	0.067	76
depth + repository + instrument + sex (cohort model)	-0.133	5.9e-07	-0.040	-0.068 to -0.013	0.0049	70

(f) The age effect as a mediation model (post hoc). Age as the exposure, the proliferation score as the mediator, the cohort's technical covariates in every equation, and a percentile bootstrap over donors. The mediator is a proxy for the counted division rate, so the indirect path is attenuated by its reliability; the second table gives the proportion mediated under a range of reliabilities.

quantity	estimate	95% CI lower	95% CI upper
total	-0.2149		
a	-0.1828		

quantity	estimate	95% CI lower	95% CI upper
b	0.8212		
direct	−0.0648	−0.1058	−0.0292
indirect	−0.1501	−0.2083	−0.0919
prop	0.6985	0.5366	0.8546

reliability	proportion mediated
1.00	0.70
0.80	0.87
0.60	1.00
0.39	1.00

Supplementary Table 4. Datasets and contrasts

All 63 contrasts, with the class assigned by experimental manipulation, the change in the cell-cycle and splicing sets, and the source publication of each series. The secretome class holds 14 contrasts.

accession	class	contrast	cell cycle	splicing
GSE14969 4	culture and metabolic	Fibroblast-D7 vs Fibroblast-D3	0.097	0.083
GSE17984 8	culture and metabolic	2-Deoxyglucose vs Control (21% O2)	0.063	−0.05
GSE17984 8	culture and metabolic	betahydroxybutyrate vs Control (21% O2)	−0.125	−0.038
GSE17984 8	culture and metabolic	Contact_Inhibition vs Control (21% O2)	−0.51	−0.184
GSE17984 8	culture and metabolic	DEX vs Control (21% O2)	0.281	0.142
GSE17984 8	culture and metabolic	Galactose vs Control (21% O2)	0.062	0.044
GSE17984 8	culture and metabolic	3% vs 21% oxygen (Control)	−0.074	−0.127
GSE17984 8	culture and metabolic	mitoNUTs vs Control (21% O2)	−0.021	−0.037
GSE17984 8	culture and metabolic	mitoNUTs+DEX vs Control (21% O2)	0.239	0.105
GSE17984 8	culture and metabolic	Oligomycin vs Control (21% O2)	0.097	0.108

accession	class	contrast	cell cycle	splicing
GSE17984 8	culture and metabolic	Oligomycin+DEX vs Control (21% O2)	0.129	0.065
GSE17984 8	culture and metabolic	SURF1 mutation vs normal (Control)	-0.334	-0.154
GSE11395 7	donor age	donor age, per decade	-0.085	-0.054
GSE30737 7	donor age	old vs young dermal fibroblast	-0.324	-0.178
GSE11696 8	photoprotection	prered_vs_uv 1h	-0.084	0.008
GSE11696 8	photoprotection	prered_vs_uv 4h	-0.061	0
GSE11696 8	photoprotection	Pre-Red vs UV, injury-residualised, 1h	-0.085	0.002
GSE11696 8	photoprotection	Pre-Red vs UV, injury-residualised, 4h	-0.063	-0.002
GSE11696 8	photoprotection	red_vs_control 1h	-0.063	-0.03
GSE11696 8	photoprotection	red_vs_control 4h	-0.058	-0.002
GSE22241 4	photoprotection	UVB + TAT-UIFSP5 vs UVB (rescue)	-0.121	-0.012
GSE22241 4	photoprotection	UVB + TAT-UIFSP6 vs UVB (rescue)	-0.027	0.048
GSE24048 6	photoprotection	0.06% Osmoter alone, no UV	-0.101	-0.101
GSE24048 6	photoprotection	0.02% Osmoter alone, no UV	0.001	-0.016
GSE24048 6	photoprotection	UV + 0.06% Osmoter vs UV (rescue)	0.076	0.024
GSE24048 6	photoprotection	UV + 0.02% Osmoter vs UV (rescue)	0.238	0.103
GSE32695 1	photoprotection	UVB + sauchinone vs UVB (rescue)	-0.035	0.175
GSE14969	reprogramming	5iLAF-D13 vs	0.066	0.053

accession	class	contrast	cell cycle	splicing
4	media	Fibroblast-D3		
GSE14969 4	reprogramming media	NHSM-D13 vs Fibroblast-D3	0.102	0.193
GSE14969 4	reprogramming media	Primed-D13 vs Fibroblast-D3	-0.021	0.011
GSE14969 4	reprogramming media	RSeT-D13 vs Fibroblast-D3	-0.051	-0.038
GSE14969 4	reprogramming media	t2iLGoY-D13 vs Fibroblast-D3	0.317	0.554
GSE29723 3	reprogramming media	O4YRSK +dox vs - dox, day 4	-0.095	-0.192
GSE29723 3	reprogramming media	OSK +dox vs -dox, day 4	0.063	-0.135
GSE13956 3	secretome	senescent-cell CM, day 4 vs day 0	0.092	0.033
GSE13956 3	secretome	senescent-cell CM, day 10 vs day 0	-0.039	-0.002
GSE25180 7	secretome	BM-MSC conditioned medium vs DMEM	0.109	0.111
GSE25180 7	secretome	BM-MSC EV- depleted medium vs DMEM	0.123	0.135
GSE25180 7	secretome	BM-MSC small EVs vs DMEM	0	0.04
GSE25180 7	secretome	BM-MSC non-small EVs vs DMEM	-0.006	0.031
GSE26605 2	secretome	cytokine-primed hMSC-CM vs untreated (both TGF- beta)	0.081	0.047
GSE26605 2	secretome	resting hMSC-CM vs untreated (both TGF- beta)	0.069	0.053
GSE26824 8	secretome	PRF serum vs untreated (gingival, boundary)	0.012	0.167
GSE27980 4	secretome	ADSC nanovesicles vs control	0.324	0.173

accession	class	contrast	cell cycle	splicing
GSE28205 4	secretome	hTSC secretome vs ES-CM	0.081	−0.129
GSE29318 6	secretome	endothelial-cell EVs vs control, 72 h	0.487	0.293
GSE30674 8	secretome	primary MSC secretome vs control	0.356	0.121
GSE30674 8	secretome	immortalised MSC secretome vs control	0.06	−0.006
GSE10970 0	senescence	deep replicative senescence	−1.906	−1.448
GSE10970 0	senescence	early replicative senescence	−1.833	−1.047
GSE17984 8	senescence	late vs early passage	−0.359	−0.252
GSE19105 5	senescence	P27 vs P4	−0.233	−0.21
GSE30695 7	senescence	DDX58_KO senescent vs proliferating	−1.21	−0.524
GSE30695 7	senescence	IFIH1_KO senescent vs proliferating	−1.373	−0.924
GSE30695 7	senescence	pLenti senescent vs proliferating	−1.297	−0.888
GSE30695 7	senescence	siScramble senescent vs proliferating	−1.073	−0.746
GSE93535	senescence	SIPS vs quiescent	−0.666	−0.409
GSE11696 8	UV injury	uv_vs_control 1h	−0.005	−0.016
GSE11696 8	UV injury	uv_vs_control 4h	−0.084	−0.082
GSE13453 3	UV injury	HDF UVB vs no UV (2 donors)	0.06	0.103
GSE22241 4	UV injury	UVB vs control (injury)	0.133	0.112
GSE24048 6	UV injury	UV vs control (injury)	−0.089	−0.052
GSE32947 5	UV injury	UVA, 3 consecutive days vs NC (injury)	0.799	0.548

Source publication of each series.

accession	source publication
GSE149694	Liu, X. et al. Nature 586, 101-107 (2020)
GSE179848	Sturm, G. et al. Sci. Data 9, 751 (2022); Commun. Biol. 6, 22 (2023)
GSE113957	Fleischer, J. G. et al. Genome Biol. 19, 221 (2018)
GSE307377	Tanaka, H. et al. bioRxiv https://doi.org/10.1101/2025.09.17.676776 (2025)
GSE116968	no associated publication
GSE222414	Yang, T. et al. MedComm 5, e625 (2024)
GSE240486	no associated publication
GSE326951	Zhang, X. et al. Pharm. Biol. 64, 764-782 (2026)
GSE297233	Lu, J. Y. et al. Cell 188, 5895-5911.e17 (2025)
GSE139563	Lopez-Antona, I. et al. Aging Cell 21, e13580 (2022)
GSE251807	Lei, R., da Silva, T. B., Cui, Z. & Ye, H. Sci. Rep. 15, 19383 (2025)
GSE266052	Brizio, M. et al. Stem Cell Res. Ther. 15, 329 (2024)
GSE268248	Imani, A., Panahipour, L., Kuhtreiber, H., Mildner, M. & Gruber, R. Cells 13, 1308 (2024)
GSE279804	Yao, T. et al. J. Cell. Mol. Med. 29, e70877 (2025)
GSE282054	Abdelmohsen, K. et al. Aging Cell 25, e70368 (2026)
GSE293186	Yuan, H. et al. J. Invest. Dermatol. 146, 1369-1381.e11 (2026)
GSE306748	no associated publication
GSE109700	De Cecco, M. et al. Nature 566, 73-78 (2019)
GSE191055	Hasegawa, T. et al. Cell 186, 1417-1431.e20 (2023)
GSE306957	Victorelli, S. et al. Nat. Commun. 16, 10992 (2025)
GSE93535	Lammermann, I. et al. npj Aging Mech. Dis. 4, 4 (2018)
GSE134533	no associated publication
GSE329475	Zhong, J. et al. Cells 15, 1497 (2026)

Intervention classes predicted out of sample (post hoc). Median absolute residual against median absolute change, from the in-sample fit of all 63 contrasts and from fits that exclude the class being predicted.

scheme	group	con tras ts	seri es	media n chang e	media n residu al	shar e predi cted	wit hin 0.0 5	bey ond 0.1 0
in-sample (63-contrast fit)	all interventions	21	10	0.111	0.038	0.65	13	7
in-sample (63-contrast fit)	reprogramming	7	2	0.135	0.115	0.14	3	4
in-sample (63-contrast fit)	secretome	14	8	0.082	0.036	0.56	10	3
leave-class- out	all interventions	21	10	0.111	0.045	0.59	12	7
leave-class- out	reprogramming	7	2	0.135	0.120	0.11	3	4
leave-class- out	secretome	14	8	0.082	0.041	0.50	9	3
all 21 interventions left out	all interventions	21	10	0.111	0.041	0.63	12	7
all 21 interventions left out	reprogramming	7	2	0.135	0.113	0.16	3	4
all 21 interventions left out	secretome	14	8	0.082	0.038	0.53	9	3
class and same-series contrasts left out	all interventions	21	10	0.111	0.045	0.59	12	7
class and same-series contrasts left out	reprogramming	7	2	0.135	0.120	0.11	3	4
class and same-series contrasts left out	secretome	14	8	0.082	0.041	0.50	9	3

Mean class residual with its 95% confidence interval and the shift detectable with 80% power.

scheme	class	unit	n	mean residual	95% CI lower	95% CI upper	detectable shift	class median change
in-samp le (63- contr ast fit)	repro gram ming	contrast	7	0.017	-0.143	0.177	0.221	0.135
in-samp le (63- contr ast fit)	repro gram ming	series	2	-0.038	-1.652	1.577	2.076	0.135
in-samp le (63- contr ast fit)	secre tome	contrast	14	-0.016	-0.062	0.030	0.065	0.082
in-samp le (63- contr ast fit)	secre tome	series	8	-0.026	-0.107	0.055	0.112	0.082
leave - class- out	repro gram ming	contrast	7	0.021	-0.140	0.182	0.222	0.135
leave - class- out	repro gram ming	series	2	-0.034	-1.654	1.587	2.083	0.135

sche me	class	unit	n	mean residua l	95% CI lower	95% CI upper	detect able shift	class media n chang e
leave - class- out	secre tome	contrast	14	−0.024	−0.070	0.022	0.065	0.082
leave - class- out	secre tome	series	8	−0.034	−0.116	0.048	0.113	0.082
all 21 inter venti ons left out	repro gram ming	contrast	7	0.015	−0.146	0.175	0.221	0.135
all 21 inter venti ons left out	repro gram ming	series	2	−0.040	−1.655	1.575	2.076	0.135
all 21 inter venti ons left out	secre tome	contrast	14	−0.019	−0.064	0.027	0.065	0.082
all 21 inter venti ons left out	secre tome	series	8	−0.028	−0.110	0.053	0.112	0.082
class and same- serie s contr asts	repro gram ming	contrast	7	0.021	−0.140	0.183	0.222	0.135

scheme	class	unit	n	mean residual	95% CI lower	95% CI upper	detectable shift	class median change
left out								
class and same-series contrasts left out	reprogramming	series	2	-0.034	-1.656	1.589	2.086	0.135
class and same-series contrasts left out	secretome	contrast	14	-0.024	-0.070	0.022	0.065	0.082
class and same-series contrasts left out	secretome	series	8	-0.034	-0.116	0.048	0.113	0.082

Supplementary Table 5. Specificity of the coupling, and the specification range of the residual age effect

All of this table is post hoc.

(a) The counted libraries. Correlation with the measured division rate across the 328 counted libraries for each splicing definition and each comparator, the partial correlation of each when the other is held constant, their difference, and the Pearson correlation between the two scores. Every set, the 177-gene set included, is scored in the expression matrix of the

programme audit so that all sets pass one filter; 176 of the 177 genes pass it, and there the 177-gene score correlates with the measured rate at 0.68 (0.71 for the score of Fig. 2a). No comparison shares a gene: shared genes were removed from both sides for the unselected definitions and from the comparator only for the selected set, and where a definition is nested in Metabolism of RNA its genes were removed from that comparator.

splicing definition	comparator	genes, splicing	genes, comp arator	rho splicing	rho com para tor	splicing comparator	comparator splicing	asymmetry	% of selected asymmetry	between scores
selected_177	translation	176	346	0.68	0.43	0.59	-0.10	0.69	100	0.68
selected_177	metabolism_of_rna	176	548	0.68	0.47	0.58	-0.22	0.80	100	0.82
selected_177	rrna_processing	176	202	0.68	0.46	0.56	-0.05	0.61	100	0.69
reactome_mrna_splicing	translation	210	348	0.63	0.43	0.51	-0.07	0.58	85	0.75
reactome_mrna_splicing	metabolism_of_rna	210	514	0.63	0.52	0.43	-0.10	0.53	66	0.89
reactome_mrna_splicing	rrna_processing	208	202	0.63	0.46	0.48	0.01	0.47	78	0.73
reactome_prem_rna_p	translation	283	343	0.65	0.43	0.55	-0.08	0.63	91	0.70

splicing definition	comparator	genes, splicing	genes, comparator	rho splicing	rho comparator	splicing comparator	comparator splicing	asymmetry	% of selected asymmetry	rank between scores
roces sing										
reactome_prem_rna_processing	metabolism_of_rna	288	436	0.65	0.46	0.54	-0.18	0.72	90	0.83
reactome_prem_rna_processing	rrna_processing	283	199	0.65	0.45	0.52	-0.01	0.52	86	0.69
mrna_splicing_not_selected	translation	84	348	0.31	0.43	0.06	0.32	-0.25	-37	0.67
mrna_splicing_not_selected	metabolism_of_rna	84	640	0.31	0.58	-0.13	0.53	-0.66	-83	0.71
mrna_splicing_not_selected	rrna_processing	83	203	0.29	0.46	0.06	0.37	-0.31	-51	0.60
prem_rna_processing_not_selected	translation	109	345	0.35	0.43	0.15	0.30	-0.15	-21	0.58

splicing definition	comparator	genes, splicing	genes, comparator	rho splicing	rho comparator	splicing comparator	comparator splicing	asymmetry	% of selected asymmetry	between scores
premrna_processing_not_selected	metabolism_of_rna	112	612	0.34	0.58	-0.07	0.50	-0.57	-72	0.70
premrna_processing_not_selected	rrna_processing	109	201	0.30	0.45	0.09	0.37	-0.27	-45	0.53

The comparisons affected by the choice of rule, repeated under the other rule: the selected set with shared genes removed from both sides, and the other definitions with shared genes removed from the comparator only.

splicing definition	comparator	shared genes removed from	shared genes	genes, splicing	genes, comparator	splicing comparator	comparator splicing	asymmetry
selected_177	translation	both sides	2	174	346	0.587	-0.099	0.686
selected_177	rrna_processing	both sides	2	174	202	0.561	-0.039	0.599
reactome_mrna_splicing	rrna_processing	comparator only	2	210	202	0.486	0.010	0.477
reactome_premrna_processing	translation	comparator only	5	288	343	0.542	-0.078	0.620

splicing definition	comparator	shared genes removed from	shared genes	genes, splicing	genes, comparator	splicing comparator	comparator splicing	asymmetry
reactome_premrna_processing	rrna_processing	comparator only	5	288	199	0.520	-0.016	0.536
mrna_splicing_not_selected	rrna_processing	comparator only	1	84	203	0.081	0.362	-0.281
premrna_processing_not_selected	translation	comparator only	3	112	345	0.114	0.304	-0.189
premrna_processing_not_selected	rrna_processing	comparator only	3	112	201	0.125	0.345	-0.219

(b) The donor cohort. Age effect per decade in the 107 adult donors under each model.

splicing definition	comparator	side	metric	model	genes	per decade	95% CI lower	95% CI upper	P
selected_177		splicing	splicing	technical covariates only	177	-0.133	-0.183	-0.084	5.9e-07
selected_177	translation	comparator	translation	technical covariates only	363	-0.067	-0.100	-0.034	0.00011
selected_177	metabolism_of_rna	comparator	metabolism_of_rna	technical covariates only	576	-0.056	-0.081	-0.030	3.3e-05

splicing definition	comparator	side	metric	model	genes	percentage	95% CI lower	95% CI upper	P
selected_177	rrna_processing	comparator	rrna_processing	technical covariates only	229	−0.074	−0.111	−0.037	0.00013
selected_177		splicing	splicing	+ proliferation score	177	−0.040	−0.068	−0.013	0.0049
selected_177	translation	comparator	translation	+ proliferation score	363	−0.032	−0.063	0.000	0.05
selected_177	metabolism_of_rna	comparator	metabolism_of_rna	+ proliferation score	576	−0.024	−0.047	−0.001	0.039
selected_177	rrna_processing	comparator	rrna_processing	+ proliferation score	229	−0.030	−0.064	0.004	0.079
selected_177	translation	splicing	splicing	+ proliferation + translation	177	−0.022	−0.043	−0.001	0.044
selected_177	translation	comparator	translation	+ proliferation + splicing	363	−0.001	−0.025	0.024	0.94
selected_177	metabolism of	splicing	splicing	+ proliferation	177	−0.016	−0.033	−0.000	0.05

splicing definition	comparator	side	metric	model	genes	percentage	95% CI lower	95% CI upper	P
	rna			n + Metabolism of RNA					
select ed_17 7	metabolism_of_ rna	comparator	metabolism_of_ rna	+ proliferation + splicing	57 6	0.00 3	-0.0 11	0.01 7	0.66
select ed_17 7	rrna_ processing	splicing	splicing	+ proliferation + rRNA processing	17 7	-0.0 22	-0.0 42	-0.0 03	0.026
select ed_17 7	rrna_ processing	comparator	rrna_ processing	+ proliferation + splicing	22 9	0.00 5	-0.0 19	0.03 0	0.68
reactome_ mrna_ splicing		splicing	splicing	technical covariates only	20 9	-0.0 94	-0.1 28	-0.0 60	2.5e- 07
reactome_ mrna_ splicing	translation	comparator	translation	technical covariates only	36 5	-0.0 68	-0.1 01	-0.0 34	0.000 11
reactome_ mrna_ splicing	metabolism_of_ rna	comparator	metabolism_of_ rna	technical covariates only	54 4	-0.0 66	-0.0 96	-0.0 37	1.7e- 05
react	rrna_	comparator	rrna_	techn	22	-0.0	-0.1	-0.0	0.000

splicing definition	comparator	side	metric	model	genes	percentage	95% CI lower	95% CI upper	P
ome_mrna_splicing	processing	or	processing	ical covariates only	9	75	12	38	13
react_ome_mrna_splicing		splicing	splicing	+ proliferation score	209	-0.042	-0.068	-0.016	0.0019
react_ome_mrna_splicing	translation	comparator	translation	+ proliferation score	365	-0.032	-0.063	-0.000	0.05
react_ome_mrna_splicing	metabolism_of_mrna	comparator	metabolism_of_mrna	+ proliferation score	544	-0.023	-0.046	0.001	0.056
react_ome_mrna_splicing	rrna_processing	comparator	rrna_processing	+ proliferation score	229	-0.030	-0.064	0.004	0.08
react_ome_mrna_splicing	translation	splicing	splicing	+ proliferation + translation	209	-0.023	-0.042	-0.005	0.015
react_ome_mrna_splicing	translation	comparator	translation	+ proliferation + splicing	365	0.004	-0.019	0.028	0.71
react_ome_mrna_splicing	metabolism_of_mrna	splicing	splicing	+ proliferation +	209	-0.020	-0.034	-0.006	0.0051

splicing definition	comparative	side	metric	model	genes	percentage	95% CI lower	95% CI upper	P
ing	rna			Metabolism of RNA					
reactome_mrna_splicing	metabolism_of_rna	comparator	metabolism_of_rna	+ proliferation + splicing	544	0.009	-0.004	0.022	0.17
reactome_mrna_splicing	rrna_processing	splicing	splicing	+ proliferation + rRNA processing	207	-0.025	-0.044	-0.007	0.0075
reactome_mrna_splicing	rrna_processing	comparator	rrna_processing	+ proliferation + splicing	229	0.009	-0.015	0.034	0.45
reactome_premrna_processing		splicing	splicing	technical covariates only	286	-0.096	-0.131	-0.061	4e-07
reactome_premrna_processing	translation	comparator	translation	technical covariates only	360	-0.068	-0.101	-0.035	0.0001
reactome_premrna_processing	metabolism_of_rna	comparator	metabolism_of_rna	technical covariates only	467	-0.061	-0.089	-0.032	4.8e-05

splicing definition	comparative side	metric	model	genes	percentage	95% CI lower	95% CI upper	P
single								
reactome_premrna_processing	comparator	rrna_processing	technical covariates only	226	-0.075	-0.112	-0.038	0.00012
reactome_premrna_processing	splicing	splicing	+ proliferation score	286	-0.037	-0.062	-0.013	0.0035
reactome_premrna_processing	translation	comparator	+ proliferation score	360	-0.032	-0.063	-0.000	0.05
reactome_premrna_processing	metabolism_of_rna	comparator	+ proliferation score	467	-0.022	-0.047	0.002	0.073
reactome_premrna_processing	rrna_processing	comparator	+ proliferation score	226	-0.030	-0.064	0.004	0.08
reactome_premrna_processing	translation	splicing	+ proliferation + translation	281	-0.021	-0.041	-0.002	0.032
reactome_premrna_processing	translation	comparator	+ proliferation	360	-0.001	-0.027	0.024	0.91

splicing definition	comparative	side	metric	model	genes	percentage	95% CI lower	95% CI upper	P
rna_processing				n + splicing					
reactome_prem_rna_processing	metabolism_of_rna	splicing	splicing	+ proliferation + Metabolism of RNA	286	-0.020	-0.035	-0.004	0.015
reactome_prem_rna_processing	metabolism_of_rna	comparator	metabolism_of_rna	+ proliferation + splicing	467	0.007	-0.009	0.022	0.42
reactome_prem_rna_processing	rrna_processing	splicing	splicing	+ proliferation + rRNA processing	281	-0.023	-0.041	-0.004	0.018
reactome_prem_rna_processing	rrna_processing	comparator	rrna_processing	+ proliferation + splicing	226	0.004	-0.022	0.031	0.75
mrna_splicing_not_selected		splicing	splicing	technical covariates only	82	-0.035	-0.060	-0.010	0.0059
mrna_splicing_not_selected	translation	comparator	translation	technical covariates only	365	-0.068	-0.101	-0.034	0.00011

splicing definition	comparator	side	metric	model	genes	percentage	95% CI lower	95% CI upper	P
mrna_splicing_not_selected	metabolism_of_rna	comparator	metabolism_of_rna	technical covariates only	671	-0.079	-0.111	-0.047	4.5e-06
mrna_splicing_not_selected	rrna_processing	comparator	rrna_processing	technical covariates only	230	-0.075	-0.112	-0.037	0.00013
mrna_splicing_not_selected		splicing	splicing	+ proliferation score	82	-0.038	-0.065	-0.011	0.0063
mrna_splicing_not_selected	translation	comparator	translation	+ proliferation score	365	-0.032	-0.063	-0.000	0.05
mrna_splicing_not_selected	metabolism_of_rna	comparator	metabolism_of_rna	+ proliferation score	671	-0.027	-0.050	-0.003	0.028
mrna_splicing_not_selected	rrna_processing	comparator	rrna_processing	+ proliferation score	230	-0.030	-0.064	0.004	0.083
mrna_splicing_not_selected	translation	splicing	splicing	+ proliferation + translation	82	-0.024	-0.048	-0.000	0.047
mrna_splicing_not_selected	translation	comparator	translation	+ proliferation +	365	-0.008	-0.037	0.020	0.55

splicing definition	comparato r	side	metric	model	genes	percentage	95% CI lower	95% CI upper	P
ected				splicing					
mrna_splicing_not_selected	metabolism_of_rna	splicing	splicing	+ proliferation + Metabolism of RNA	82	-0.018	-0.040	0.003	0.089
mrna_splicing_not_selected	metabolism_of_rna	comparator	metabolism_of_rna	+ proliferation + splicing	671	-0.005	-0.024	0.014	0.6
mrna_splicing_not_selected	rrna_processing	splicing	splicing	+ proliferation + rRNA processing	81	-0.026	-0.050	-0.002	0.035
mrna_splicing_not_selected	rrna_processing	comparator	rrna_processing	+ proliferation + splicing	230	-0.007	-0.038	0.024	0.67
prem_rna_processing_not_selected		splicing	splicing	technical covariates only	109	-0.034	-0.057	-0.011	0.0043
prem_rna_processing_	translation	comparator	translation	technical covariates	362	-0.068	-0.102	-0.035	9.8e-05

splicing definition	comparative	side	metric	model	genes	percentage	95% CI lower	95% CI upper	P
not_selected				only					
prem_rna_processes_not_selected	metabolism_of_rna	comparator	metabolism_of_rna	technical_covariates_only	644	-0.081	-0.114	-0.048	4.3e-06
prem_rna_processes_not_selected	rrna_processing	comparator	rrna_processing	technical_covariates_only	228	-0.075	-0.112	-0.038	0.00012
prem_rna_processes_not_selected		splicing	splicing	+ proliferation score	109	-0.032	-0.058	-0.007	0.013
prem_rna_processes_not_selected	translation	comparator	translation	+ proliferation score	362	-0.032	-0.063	-0.000	0.05
prem_rna_processes_not_selected	metabolism_of_rna	comparator	metabolism_of_rna	+ proliferation score	644	-0.027	-0.051	-0.003	0.028
prem_rna_processes_not_selected	rrna_processing	comparator	rrna_processing	+ proliferation	228	-0.030	-0.065	0.004	0.08

splicing definition	comparative	side	metric	model	genes	percentage	95% CI lower	95% CI upper	P
proteins sing_ not_ selected	ssing		ssing	eration score					
premature proteins sing_ not_ selected	translation	splicing	splicing	+ proliferation + translation	106	-0.021	-0.045	0.003	0.084
premature proteins sing_ not_ selected	translation	comparator	translation	+ proliferation + splicing	362	-0.014	-0.043	0.016	0.35
premature proteins sing_ not_ selected	metabolism_of_rna	splicing	splicing	+ proliferation + Metabolism of RNA	109	-0.014	-0.034	0.006	0.17
premature proteins sing_ not_ selected	metabolism_of_rna	comparator	metabolism_of_rna	+ proliferation + splicing	644	-0.007	-0.026	0.012	0.46
premature proteins sing_ not_ selected	rrna_processing	splicing	splicing	+ proliferation + rRNA processing	106	-0.022	-0.045	0.001	0.057

splicing definition	comparator	side	metric	model	genes	percentage	95% CI lower	95% CI upper	P
selected				splicing					
prem_rna_processing_not_selected	rrna_processing	comparator	rrna_processing	+ proliferation + splicing	228	-0.010	-0.042	0.021	0.52

(c) GTEx cultures. The same comparison against the 20-marker transcriptional proliferation score in 511 cultures. The asymmetry is not reproduced.

splicing definition	comparator	n	rho splicing	rho comparator	splicing comparator	comparator splicing	asymmetry
selected_177	translation	511	0.75	0.78	0.41	0.51	-0.10
selected_177	metabolism_of_rna	511	0.75	0.62	0.60	-0.29	0.88
selected_177	rrna_processing	511	0.75	0.66	0.53	0.26	0.27
reactome_mrna_splicing	translation	511	0.68	0.78	0.28	0.57	-0.29
reactome_mrna_splicing	metabolism_of_rna	511	0.68	0.66	0.25	0.10	0.14
reactome_mrna_splicing	rrna_processing	511	0.68	0.66	0.41	0.35	0.06
reactome_pre_mrna_processing	translation	511	0.68	0.78	0.29	0.59	-0.30
reactome_pre_mrna_processing	metabolism_of_rna	511	0.68	0.66	0.28	0.11	0.16
reactome_pre_mrna_processing	rrna_processing	511	0.68	0.65	0.40	0.34	0.06
mrna_splicing	translation	511	0.43	0.78	0.00	0.72	-0.72

splicing definition	comparator	n	rho splici ng	rho com para tor	splici ng comp arator	comp arator splici ng	asym metry
_not_selected							
mrna_splicing _not_selected	metabolism_o f_rna	511	0.43	0.70	-0.44	0.70	-1.14
mrna_splicing _not_selected	rrna_processi ng	511	0.42	0.66	0.07	0.57	-0.50
premrna_proc essing_not_se lected	translation	511	0.40	0.78	0.01	0.73	-0.73
premrna_proc essing_not_se lected	metabolism_o f_rna	511	0.41	0.71	-0.54	0.76	-1.31
premrna_proc essing_not_se lected	rrna_processi ng	511	0.38	0.65	0.01	0.58	-0.56

(d) The two cohorts, not pooled. Proliferation-adjusted age effects of the two fibroblast cohorts, reported side by side; they are not combined, because one is an attenuation and the other a suppression effect in cultures with no unadjusted age effect.

gene set	cohort	n	per decade	95% CI lower	95% CI upper	P
preregistere d 96-gene set	GSE113957 fibroblast donors	107	-0.0330	-0.0644	-0.0016	0.04
preregistere d 96-gene set	GTEx cultured fibroblasts	492	-0.0529	-0.0866	-0.0192	0.0022
Reactome pre-mRNA processing	GSE113957 fibroblast donors	107	-0.0373	-0.0620	-0.0126	0.0035
Reactome pre-mRNA processing	GTEx cultured fibroblasts	492	-0.0448	-0.0729	-0.0167	0.0018

(e) The four definitions of the fibroblast cohort, reported as sensitivity and not pooled, since they are not independent of one another.

metric	cohort	n	adjusted per decade	95% CI lower	95% CI upper	P
machinery	adult2082	76	-0.056	-0.095	-0.017	0.0057
machinery	normal	133	-0.045	-0.069	-0.020	0.00044
machinery	primary	107	-0.040	-0.068	-0.013	0.0049
machinery	published	142	-0.023	-0.044	-0.002	0.035
machinery96	adult2082	76	-0.068	-0.115	-0.021	0.0053
machinery96	normal	133	-0.058	-0.085	-0.032	3.3e-05
machinery96	primary	107	-0.033	-0.064	-0.002	0.04
machinery96	published	142	-0.031	-0.053	-0.009	0.0069

(f) Every sample-level step in one unit (post hoc). The 20-marker proliferation score calibrated against the counted division rate in the counted libraries, so that each dataset's slope can be read in divisions per day. Score-based slopes are multiplied by 0.51, the factor by which the naive conversion overstates the effect where counting is available.

dataset	n	per SD of proliferation	per div. per day	rho
counted resource, against counting	328	0.424	2.90	0.705
counted resource, against the score	328	0.830	2.90	0.842
donor cohort	107	0.891	3.11	0.912
GTEEx cultured fibroblasts	492	0.738	2.58	0.739
GTEEx skin, sun-exposed	739	0.247	0.86	0.225
GTEEx skin, not exposed	626	0.108	0.38	0.056

Supplementary Table 6. Ageing programmes against measured division rate and donor age

First: correlation with the measured division rate across 328 counted libraries, with all genes and after removing every gene belonging to the Reactome cell-cycle union. Second: the same programmes against donor age in the 107-donor primary cohort, before and after adjustment for proliferation (adj), each with its nominal P value and BH-FDR.

programme	genes	cell-cycle genes removed	rho, all genes	rho, after removal	FDR
chromatin organisation	312	38	0.69	0.63	1.3e-36
cellular senescence	150	82	0.67	0.38	1.6e-12
pre-mRNA processing	287	46	0.65	0.67	5.9e-42
mRNA splicing	198	12	0.63	0.64	8.5e-39
cell cycle (positive control)	499	499	0.62		
interferon signalling	279	67	0.60	0.49	1.7e-20
DNA double-strand repair	180	124	0.59	0.55	1.7e-26
telomere maintenance	82	82	0.58		
base excision repair	63	44	0.56	0.35	2.1e-10
interleukin signalling	334	65	0.47	0.26	2.8e-06
rRNA processing	187	9	0.46	0.45	5.5e-17
mitochondrial translation	100	0	0.45	0.45	1.4e-17
translation	348	39	0.43	0.42	1.2e-14
extracellular matrix	235	1	-0.33	-0.34	7.4e-10
TCA cycle	34	0	0.29	0.29	9.3e-08
insulin / IGF signalling	62	3	-0.27	-0.31	2e-08
respiratory electron transport	142	0	0.22	0.22	5.7e-05
proteasome	268	64	0.22	-0.08	0.16
chaperone / HSF1	30	5	0.21	0.10	0.073
NF-kB / TNF	32	5	-0.14	-0.23	3.9e-05
autophagy	130	38	0.12	-0.03	0.65
mTOR signalling	23	0	0.11	0.11	0.062
unfolded protein response	89	3	0.05	0.05	0.4
collagen formation	71	0	-0.02	-0.02	0.76

program me	rho with age	per decade	P	FDR	per decade (adj)	P (adj)	FDR (adj)
chromati	-0.51	-0.038	1.7e-05	8.3e-05	-0.015	0.037	0.25

program me	rho with age	per decade	P	FDR	per decade (adj)	P (adj)	FDR (adj)
n organisat ion							
cellular senescen ce	−0.37	−0.026	0.0024	0.0044	0.000	1	1
pre- mRNA processin g	−0.49	−0.096	3.8e-07	4.6e-06	−0.037	0.0034	0.04
mRNA splicing	−0.51	−0.098	2.1e-07	4.6e-06	−0.043	0.0017	0.04
cell cycle (positive control)	−0.47	−0.088	2.9e-05	0.0001	−0.006	0.42	0.64
interfero n signalling	−0.30	−0.027	0.0024	0.0044	−0.006	0.46	0.64
DNA double- strand repair	−0.47	−0.099	9.1e-06	7.3e-05	−0.014	0.11	0.35
telomere maintena nce	−0.49	−0.096	1.4e-05	8.3e-05	−0.011	0.19	0.38
base excision repair	−0.50	−0.097	2.3e-05	9.1e-05	−0.009	0.3	0.51
interleuki n signalling	−0.19	−0.006	0.3	0.33	0.002	0.78	0.94
rRNA processin g	−0.42	−0.077	0.00017	0.0004	−0.038	0.052	0.25
mitochon drial translatio n	−0.42	−0.082	9.5e-05	0.00028	−0.022	0.18	0.38
translatio	−0.46	−0.068	0.00011	0.00029	−0.032	0.05	0.25

program me	rho with age	per decade	P	FDR	per decade (adj)	P (adj)	FDR (adj)
n							
TCA cycle	−0.18	−0.020	0.19	0.25	0.007	0.66	0.88
respirator y electron transport	−0.28	−0.018	0.19	0.25	−0.001	0.93	0.98
proteaso me	−0.01	0.011	0.093	0.14	0.009	0.22	0.4
chaperon e / HSF1	−0.23	−0.008	0.55	0.57	0.011	0.38	0.61
autophag y	−0.07	−0.001	0.95	0.95	−0.003	0.78	0.94
mTOR signalling	−0.14	−0.012	0.31	0.33	0.001	0.94	0.98
unfolded protein response	−0.43	−0.020	0.044	0.07	−0.016	0.13	0.35
collagen formation	−0.44	−0.015	0.21	0.25	−0.020	0.13	0.35
NF-kB / TNF	0.13	0.028	0.0023	0.0044	0.014	0.12	0.35
insulin / IGF signalling	0.08	0.020	0.025	0.044	0.012	0.18	0.38
extracellu lar matrix	−0.18	0.013	0.15	0.21	−0.002	0.85	0.98

Supplementary Table 7. Senescence panels in cultured fibroblasts

Post hoc. Cell-cycle content is the share of a panel's measured genes belonging to the Reactome cell-cycle union. Correlations are with the measured division rate across 328 counted libraries; age effects are per decade in the 107 adult donors. The percentage of the age effect lost is meaningful only for the five panels that had one.

panel	measured	% cell cycle	rho rate	rho after removal	median gene rho	age per decade	FDR	adjusted	FD R adj	% of age effect lost
Reactome Cellular Senescence	150	55	0.67	0.38	0.21	-0.026	0.0041	0.000	1	100
SenMayo	88	1	-0.10	-0.14	-0.05	0.003	0.72	0.002	1	-
Fridman senescence, up	71	15	-0.07	0.01	-0.07	0.043	0.00039	0.015	0.93	66
Fridman senescence, down	11	27	0.38	0.01	0.16	-0.034	0.02	-0.005	1	87
CellAge, induces senescence	315	10	0.33	0.21	0.02	0.002	0.72	0.004	1	-
CellAge, inhibits senescence	440	21	0.69	0.71	0.20	-0.041	0.00037	-0.002	1	95
Reactome Cell Cycle	499	100	0.62		0.34	-0.088	0.00021	-0.006	1	93

panel	me asu red	% cell cyc le	rho rate	rho after remo val	medi an gene rho	age per decad e	FDR	adjust ed	FD R adj	% of ag e eff ect lost
, Mitotic (control)										

Supplementary Table 8. Preregistered methylation clock tests

The four hypotheses fixed in Supplementary Note 2, each with the criterion it was given. M1 and M2 are Spearman correlations across methylation samples; M3 and M4 are across the 109 cultures carrying both a clock and transcriptional scores. “Met” is the preregistered criterion, which for M1 to M3 is $|\text{rho}| \geq 0.3$ with a significant P value and for M4 is an attenuation of at least 50% when the measured division rate is held constant.

test	rho	P	FDR	n	met
M1 DNAm age vs cumulative doublings (untreated healthy)	0.356	0.00077	0.0015	86	TRUE
M2 clock acceleration vs measured division rate	−0.014	0.76	0.76	471	FALS E
M3 proliferation score vs DNAm age (paired cultures)	0.162	0.093	0.12	109	FALS E
M4 splicing vs DNAm age, division rate held constant	−0.338	0.00032	0.0013	109	FALS E

Supplementary Table 9. The preregistered GTEx hypotheses and their outcomes

The six hypotheses as they were locked before the data were downloaded, with the criterion fixed in advance, the value observed and the verdict. Four were supported; H4 could not be evaluated because the GTEx cultures had no unadjusted age effect to attenuate; and H6 — a positive control asking whether collagen formation falls with age in sun-exposed skin — was not supported, which under the preregistered rule bars age effects measured within GTEx tissues from supporting any claim in this paper. The full

document, its English translation and its ten recorded deviations are Supplementary Note 1.

hypothesis	test	criterion	observed	verdict
H1	culture: splicing genes track proliferation	$\rho \geq 0.5$	0.74	supported
H2	skin: coupling weaker than in culture	one-sided P < 0.05	0.23 vs 0.74	supported
H3	skeletal muscle: no coupling	$\rho < 0.3$	-0.12	supported
H4	culture: age effect shrinks with adjustment	$\geq 50\%$ smaller	no unadjusted effect (P = 0.33)	not evaluable (no unadjusted age effect)
H5	culture: collagen formation independent of proliferation	$ \rho < 0.3$	0.05	supported
H6	skin: collagen formation falls with age	negative, P < 0.05	+0.027 per decade (P = 0.018)	not supported

Supplementary Table 10. Programmes against proliferation and donor age in GTEx

(a) Partial correlation with proliferation. Partial Spearman correlation of each of the twenty-five programme scores with the proliferation score in each tissue, after regressing out RNA integrity, ischaemic time and library size (the values of Fig. 8b).

programme	cultured fibroblasts	skin, sun- exposed	skin, not exposed	skeletal muscle
splicing 96	0.74	0.23	0.06	-0.12
pre-mRNA processing	0.69	0.14	-0.02	-0.13
mRNA splicing	0.69	0.13	-0.04	-0.14
translation	0.75	0.21	0.09	-0.21
rRNA processing	0.70	0.18	0.01	-0.11

programme	cultured fibroblasts	skin, sun-exposed	skin, not exposed	skeletal muscle
proteasome	0.27	0.16	0.02	−0.37
autophagy	0.44	0.06	−0.10	−0.33
chaperone / HSF1	0.17	0.21	0.07	−0.31
unfolded protein response	0.70	0.03	−0.08	−0.16
DNA double-strand repair	0.90	0.64	0.58	−0.05
base excision repair	0.85	0.62	0.58	0.30
telomere maintenance	0.81	0.61	0.52	0.09
chromatin organisation	0.43	0.14	0.02	−0.31
mitochondrial translation	0.42	0.23	0.13	−0.15
respiratory electron transport	0.67	0.26	0.24	−0.12
TCA cycle	0.74	0.19	0.18	−0.10
interferon signalling	0.59	0.24	0.06	0.01
NF-κB / TNF	−0.56	0.04	−0.12	−0.36
extracellular matrix	−0.21	−0.10	−0.01	0.22
collagen formation	0.05	0.03	0.09	0.24
insulin / IGF signalling	−0.08	−0.11	−0.11	−0.07
mTOR signalling	0.41	0.13	−0.01	−0.31
cellular senescence	0.63	0.19	0.05	−0.21
interleukin signalling	0.42	−0.02	−0.09	−0.13
cell cycle (positive control)	0.94	0.67	0.60	0.03

(b) Age effects before and after adjustment for proliferation. Change in each programme score per decade of donor age, from linear models with sex, Hardy death classification, RNA integrity, ischaemic time and library size as covariates, and the same with the proliferation score added (adj). FDR is Benjamini-Hochberg within a tissue and was computed for the unadjusted effects. The preregistered positive control for age effects in tissue failed (H6), so under the preregistered rule the age effects in skin and muscle are listed for completeness and support no claim.

Cultured fibroblasts (492 donors).

programme	per decade	P	FDR	per decade (adj)	P (adj)
splicing 96	−0.025	0.33	0.51	−0.053	0.0022
pre-mRNA processing	−0.025	0.2	0.41	−0.045	0.0018
mRNA splicing	−0.026	0.21	0.41	−0.047	0.0023
translation	0.004	0.81	0.88	−0.012	0.32
rRNA processing	0.006	0.72	0.88	−0.009	0.43
proteasome	−0.023	0.00085	0.02	−0.025	0.00017
autophagy	−0.015	0.047	0.23	−0.020	0.0046
chaperone / HSF1	−0.013	0.13	0.33	−0.016	0.069
unfolded protein response	−0.007	0.54	0.8	−0.019	0.03
DNA double-strand repair	−0.000	0.99	0.99	−0.027	0.0027
base excision repair	−0.001	0.95	0.99	−0.027	0.014
telomere maintenance	−0.011	0.63	0.87	−0.037	0.0044
chromatin organisation	−0.021	0.036	0.23	−0.026	0.0034
mitochondrial translation	−0.036	0.075	0.27	−0.047	0.011
respiratory electron transport	−0.005	0.77	0.88	−0.021	0.18
TCA cycle	0.018	0.31	0.51	0.000	0.98
interferon signalling	−0.002	0.74	0.88	−0.006	0.17
NF-kB / TNF	−0.034	0.0016	0.02	−0.024	0.0042
extracellular matrix	0.014	0.12	0.33	0.017	0.063
collagen formation	0.017	0.12	0.33	0.015	0.16
insulin / IGF signalling	−0.008	0.19	0.41	−0.007	0.23
mTOR signalling	−0.011	0.3	0.51	−0.016	0.12
cellular senescence	−0.014	0.069	0.27	−0.022	0.0004
interleukin signalling	−0.010	0.025	0.21	−0.013	0.0013
cell cycle (positive control)	0.006	0.75	0.88	−0.021	0.0018

Skin, sun-exposed (739 donors).

programme	per decade	P	FDR	per decade (adj)	P (adj)
splicing 96	−0.032	0.037	0.085	−0.024	0.1
pre-mRNA processing	−0.014	0.2	0.24	−0.010	0.33
mRNA splicing	−0.012	0.31	0.35	−0.009	0.46
translation	−0.046	0.0024	0.029	−0.038	0.0086

programme	per decade	P	FDR	per decade (adj)	P (adj)
rRNA processing	−0.037	0.03	0.075	−0.029	0.079
proteasome	−0.012	0.072	0.12	−0.009	0.14
autophagy	−0.025	0.0042	0.031	−0.023	0.0078
chaperone / HSF1	−0.029	0.013	0.051	−0.025	0.029
unfolded protein response	−0.012	0.12	0.15	−0.012	0.13
DNA double-strand repair	−0.016	0.077	0.12	−0.005	0.49
base excision repair	−0.006	0.57	0.57	0.006	0.44
telomere maintenance	−0.018	0.068	0.12	−0.006	0.42
chromatin organisation	−0.016	0.016	0.051	−0.014	0.034
mitochondrial translation	−0.043	0.0015	0.029	−0.037	0.0056
respiratory electron transport	−0.026	0.063	0.12	−0.020	0.15
TCA cycle	−0.025	0.078	0.12	−0.020	0.16
interferon signalling	−0.007	0.33	0.35	−0.003	0.67
NF-κB / TNF	−0.016	0.12	0.15	−0.014	0.17
extracellular matrix	0.029	0.0067	0.033	0.027	0.014
collagen formation	0.027	0.018	0.051	0.027	0.019
insulin / IGF signalling	−0.011	0.089	0.13	−0.012	0.059
mTOR signalling	−0.029	0.018	0.051	−0.025	0.04
cellular senescence	−0.006	0.2	0.24	−0.004	0.38
interleukin signalling	0.006	0.34	0.35	0.007	0.27
cell cycle (positive control)	−0.023	0.005	0.031	−0.012	0.044

Skin, not exposed (626 donors).

programme	per decade	P	FDR	per decade (adj)	P (adj)
splicing 96	−0.023	0.16	0.78	−0.024	0.14
pre-mRNA processing	−0.001	0.93	0.97	−0.001	0.91
mRNA splicing	−0.002	0.84	0.95	−0.003	0.83
translation	−0.014	0.37	0.78	−0.015	0.32
rRNA processing	−0.025	0.18	0.78	−0.026	0.16
proteasome	0.002	0.71	0.91	0.002	0.73
autophagy	−0.013	0.16	0.78	−0.013	0.16

programme	per decade	P	FDR	per decade (adj)	P (adj)
chaperone / HSF1	−0.023	0.09	0.78	−0.023	0.083
unfolded protein response	−0.018	0.053	0.78	−0.017	0.057
DNA double-strand repair	0.008	0.37	0.78	0.005	0.45
base excision repair	0.008	0.46	0.78	0.005	0.58
telomere maintenance	−0.003	0.76	0.91	−0.006	0.48
chromatin organisation	−0.007	0.35	0.78	−0.007	0.33
mitochondrial translation	−0.013	0.37	0.78	−0.014	0.31
respiratory electron transport	0.006	0.69	0.91	0.004	0.77
TCA cycle	0.013	0.4	0.78	0.011	0.46
interferon signalling	0.001	0.95	0.97	0.000	0.99
NF-κB / TNF	−0.005	0.65	0.91	−0.005	0.68
extracellular matrix	−0.006	0.63	0.91	−0.006	0.65
collagen formation	−0.011	0.38	0.78	−0.011	0.38
insulin / IGF signalling	−0.008	0.26	0.78	−0.008	0.28
mTOR signalling	−0.015	0.26	0.78	−0.015	0.24
cellular senescence	−0.004	0.47	0.78	−0.004	0.43
interleukin signalling	−0.000	0.97	0.97	−0.000	0.99
cell cycle (positive control)	0.003	0.77	0.91	−0.000	0.96

Skeletal muscle (839 donors).

programme	per decade	P	FDR	per decade (adj)	P (adj)
splicing 96	0.012	0.31	0.54	0.016	0.19
pre-mRNA processing	0.008	0.41	0.64	0.011	0.22
mRNA splicing	0.007	0.48	0.7	0.011	0.25
translation	−0.037	0.0018	0.0075	−0.031	0.0097
rRNA processing	−0.016	0.25	0.49	−0.011	0.42
proteasome	−0.023	0.00024	0.0022	−0.017	0.0051
autophagy	−0.022	0.0033	0.0092	−0.015	0.032
chaperone / HSF1	−0.011	0.32	0.54	0.002	0.88
unfolded protein response	−0.010	0.15	0.31	−0.008	0.26
DNA double-strand repair	−0.000	0.98	0.99	−0.001	0.86

programme	per decade	P	FDR	per decade (adj)	P (adj)
base excision repair	0.020	0.012	0.029	0.012	0.11
telomere maintenance	0.001	0.86	0.99	-0.002	0.76
chromatin organisation	-0.000	0.99	0.99	0.005	0.37
mitochondrial translation	-0.053	0.00062	0.0039	-0.046	0.003
respiratory electron transport	-0.049	0.0029	0.0091	-0.042	0.011
TCA cycle	-0.036	0.034	0.077	-0.029	0.084
interferon signalling	0.000	0.96	0.99	0.000	0.94
NF- κ B / TNF	-0.024	0.00026	0.0022	-0.017	0.0065
extracellular matrix	0.029	0.0019	0.0075	0.024	0.011
collagen formation	0.033	0.0021	0.0075	0.026	0.015
insulin / IGF signalling	-0.000	0.99	0.99	-0.000	0.98
mTOR signalling	-0.037	0.00027	0.0022	-0.029	0.0033
cellular senescence	-0.000	0.93	0.99	0.002	0.66
interleukin signalling	-0.002	0.75	0.99	-0.001	0.91
cell cycle (positive control)	-0.003	0.59	0.83	-0.005	0.31

(c) Range restriction (post hoc). Spread of the proliferation score, the partial correlation of the splicing score with it, and the regression slope of the splicing score on it with its 95% confidence interval, which a narrower spread does not affect; for the preregistered 96-gene set and for the 177-gene set.

tissue (donors)	genes in set	SD of proliferation	partial rho	P	slope	95% CI of slope
cultured fibroblasts (492)	96	0.956	0.739	1.2e-85	0.554	0.509 to 0.599
cultured fibroblasts (492)	177	0.956	0.766	1.3e-95	0.569	0.526 to 0.611
skin, sun-exposed (739)	96	0.752	0.225	6.6e-10	0.167	0.120 to 0.215
skin, sun-exposed (739)	177	0.752	0.203	2.9e-08	0.118	0.079 to 0.156
skin, not exposed (626)	96	0.743	0.056	0.17	0.072	0.020 to 0.125

tissue (donors)	genes in set	SD of proliferation	partial rho	P	slope	95% CI of slope
skin, not exposed (626)	177	0.743	0.021	0.6	0.037	−0.005 to 0.079
skeletal muscle (839)	96	0.397	−0.115	0.00085	−0.127	−0.203 to −0.051
skeletal muscle (839)	177	0.397	−0.130	0.00016	−0.139	−0.206 to −0.072

The partial correlation expected in each tissue if the narrower spread of proliferation were the only difference from culture, against the one observed (96-gene set).

tissue	SD relative to culture	rho expected from range restriction alone	rho observed
skin, sun-exposed	0.786	0.653	0.225
skin, not exposed	0.776	0.649	0.056
skeletal muscle	0.415	0.415	−0.115

Cultured fibroblasts subsampled to the spread of proliferation in sun-exposed skin: a window centred on the median, and kernel weights that down-weight the tails without discarding them; the median partial correlation over the draws with its 2.5th and 97.5th percentiles.

scheme	target SD	cultures available	draws	SD achieved (median)	partial rho (median)	2.5th to 97.5th percentile
window	0.752	338	500	0.751	0.872	0.834 to 0.910
kernel	0.752	492	500	0.801	0.770	0.701 to 0.857

(d) Adjustment for cell composition (post hoc). The splicing-proliferation partial correlation with the technical covariates only, and with the marker scores named in the third column added to them. Block (f) repeats the adjustment across programmes, which is what separates a composition effect on splicing from a composition effect on the whole proliferation axis.

tissue (donors)	genes in set	marker scores added	partial rho, technical	P	partial rho, with composition	P (composition)
cultured fibroblasts (492)	96	fibroblast, immune	0.739	1.2e-85	0.722	1.4e-79
cultured fibroblasts (492)	177	fibroblast, immune	0.766	1.3e-95	0.764	3.2e-94
skin, sun- exposed (739)	96	keratinocyte, fibroblast, immune	0.225	6.6e-10	0.101	0.0064
skin, sun- exposed (739)	177	keratinocyte, fibroblast, immune	0.203	2.9e-08	0.123	0.00088
skin, not exposed (626)	96	keratinocyte, fibroblast, immune	0.056	0.17	-0.058	0.15
skin, not exposed (626)	177	keratinocyte, fibroblast, immune	0.021	0.6	-0.047	0.25
skeletal muscle (839)	96	myofibre, satellite cell, fibroblast, immune	-0.115	0.00085	-0.040	0.25
skeletal muscle (839)	177	myofibre, satellite cell, fibroblast, immune	-0.130	0.00016	-0.056	0.11

(e) Positive control for donor concordance (post hoc). Spearman correlation between a donor's cultured fibroblasts and the same donor's sun-exposed skin for genes whose expression depends strongly on genotype or on sex. The column before the note is the product of the gene's correlations with sex in the two sample types, that is, the concordance that sex alone would produce. For comparison, the preregistered exploratory analysis gave $\rho = -0.10$ to $+0.13$ for the 25 programme scores in the same 388 donors (Supplementary Fig. S8c).

gene	group	don ors	rho	P	rho expec ted from sex alone	note
XIST	sex chromosome	388	0.730	1e-65	0.664	
RPS4Y1	sex chromosome	388	0.666	5.5e-51	0.670	
DDX3Y	sex chromosome	388	0.729	1.8e-65	0.671	
KDM5D	sex chromosome	388	0.692	1.5e-56	0.670	
UTY	sex chromosome	388	0.668	1.6e-51	0.666	
USP9Y	sex chromosome	388	0.698	5.8e-58	0.667	
EIF1AY	sex chromosome	388	0.736	1.7e-67	0.668	
GSTM1	genotype- dependent	388	0.750	2.9e-71	0.004	
GSTT1	genotype- dependent					absent from the gene- symbol map
UGT2B1 7	genotype- dependent					present in the annotation but filtered out as not expressed
ERAP2	genotype- dependent	388	0.749	5.1e-71	0.002	

(f) The same composition adjustment for six sets (post hoc). Run on the splicing sets alone, block (d) cannot distinguish an adjustment that removes splicing's coupling from one that flattens the proliferation signal. Repeated for the mitotic cell-cycle positive control, collagen formation and the two unselected Reactome splicing definitions, the two separate: the cell-cycle programme and collagen formation do not move, while every definition of the splicing set falls (the values of Fig. 8d). The fibroblast marker score contains COL1A1 and COL1A2, so collagen formation is not an independent check.

tissue	programme	partial rho, technical	partial rho, with composition	difference
cultured fibroblasts	96	0.739	0.722	0.017
cultured fibroblasts	177	0.766	0.764	0.003
cultured fibroblasts	pre-mRNA processing	0.691	0.653	0.038
cultured fibroblasts	mRNA splicing	0.689	0.651	0.038
cultured fibroblasts	collagen formation	0.049	0.422	-0.373
cultured fibroblasts	cell cycle (positive control)	0.941	0.952	-0.011
skin, sun- exposed	96	0.225	0.101	0.124
skin, sun- exposed	177	0.203	0.123	0.080
skin, sun- exposed	pre-mRNA processing	0.145	0.082	0.062
skin, sun- exposed	mRNA splicing	0.134	0.063	0.071
skin, sun- exposed	collagen formation	0.030	0.027	0.003
skin, sun- exposed	cell cycle (positive control)	0.670	0.666	0.004
skin, not exposed	96	0.056	-0.058	0.114
skin, not exposed	177	0.021	-0.047	0.068
skin, not exposed	pre-mRNA processing	-0.025	-0.074	0.049
skin, not exposed	mRNA splicing	-0.044	-0.094	0.049
skin, not exposed	collagen formation	0.086	0.110	-0.024
skin, not exposed	cell cycle (positive	0.597	0.586	0.011

tissue	programme	partial rho, technical	partial rho, with composition	difference
	control)			
skeletal muscle	96	-0.115	-0.040	-0.075
skeletal muscle	177	-0.130	-0.056	-0.074
skeletal muscle	pre-mRNA processing	-0.131	-0.026	-0.105
skeletal muscle	mRNA splicing	-0.139	-0.037	-0.102
skeletal muscle	collagen formation	0.240	0.031	0.210
skeletal muscle	cell cycle (positive control)	0.031	0.157	-0.126

(g) What a two-compartment tissue should show (post hoc). Between-donor variation in each score written as a part that follows the fraction of cycling cells and a part that does not. The proliferation score and the mitotic cell-cycle programme are both read from cycling cells, so their correlation fixes how well the design sees that fraction; if a share w of the splicing score's variation followed it, the expected correlation would be w times that value. The row without a w is the observed pair.

share w	expected splicing rho	expected cell-cycle rho
0.10	0.067	0.667
0.20	0.131	0.665
0.30	0.194	0.667
0.40	0.261	0.663
0.50	0.329	0.666
0.60	0.394	0.666
0.70	0.460	0.668
0.80	0.528	0.667
0.90	0.597	0.665
1.00	0.666	0.668
	0.225	0.670

Supplementary Note 1. Preregistration for the GTEx analysis

What this document is

The GTEx analysis reported in Fig. 8 was planned before the data were seen. The plan below was written in Korean on 13 September 2026, and its SHA-256 digest was recorded in `public_data_tierA/logs/preregistration_sha256.txt` before any GTEx expression file was downloaded or opened; the analysis script `scripts/run_gtex_boundary.R` recomputes the digest and stops if it does not match. The digest is

61972b991dc6de770478cd7f6df39a5319e11107ef2993aac2faeee8eb8b774e

and it still matches the current file, which has not been amended since it was locked.

This was an internal record. It was **not** deposited with a public registry such as OSF or AsPredicted, so a reader cannot verify independently when it was written; what can be verified is that the document the analysis ran against is the document reproduced here, and that its stated criteria are the criteria against which the outcomes below are reported, including the two that the analysis failed. The translation is complete rather than summarised. Gene names, dataset identifiers, statistical terms and numbers are unchanged; where the Korean is ambiguous it is translated literally and a bracketed translator's note is added.

Translation

Preregistration: culture-versus-tissue boundary test (GTEx v8 via recount3)

- Written: 2026-09-13
- Status: **written and locked before the GTEx expression data were downloaded or opened**
- Locking method: record the SHA-256 of this file in `public_data_tierA/logs/preregistration_sha256.txt`, then run the analysis script.

1. Purpose

The manuscript's present claim is as follows. In cultured human fibroblasts, splicing-factor expression and eight ageing programmes (chromatin organisation, cellular senescence, pre-mRNA processing, mRNA splicing, interferon signalling, DNA double-strand break repair, telomere

maintenance, base excision repair) are readouts of division rate, whereas collagen formation and the unfolded protein response are not. The limitations section stated that “it says nothing about post-mitotic tissue”. This test establishes that boundary from data rather than by assumption. GTEx is the only public resource in which **cultured fibroblasts**, **skin tissue** and **skeletal muscle (post-mitotic tissue)** can be compared under one definition in the same donor population.

2. Data (processed only)

File	Content	Size
gtex.gene_sums.SKIN .G026.gz + 3 metadata files	Cells - Cultured fibroblasts, Skin - Sun Exposed (Lower leg), Skin - Not Sun Exposed (Suprapubic)	about 196 MB
gtex.gene_sums.MUSC LE.G026.gz + 3 metadata files	Muscle - Skeletal	about 81 MB

About 277 MB in total. FASTQ and junction files will not be downloaded. Splicing **outcome** (unannotated junctions) will therefore not be computed in GTEx, and only programme-level expression analysis will be performed.

3. Definitions fixed before seeing results

- **Gene ID → symbol:** use the mapping already used in the project (Geneid → Gene_name from GSE282054_raw_counts.txt.gz), unchanged.
- **Normalisation:** separately per tissue type. edgeR filterByExpr → TMM → log₂ CPM → gene-wise z.
- **Programme scores:** the list of 24 Reactome sets in scripts/run_hallmark_audit.R, unchanged. Mean z of the genes expressed in that tissue; excluded if fewer than 15.
- **Splicing set:** the 96 genes in conserved_core/age_down_splicing_core.txt.
- **Proliferation score:** mean z of the 20 markers in scripts/run_outcome_vs_expression.R (MKI67, CCNB1, CCNA2, CDK1, TOP2A, BIRC5, BUB1, PLK1, AURKA, TYMS, RRM2, PCNA, MCM2-MCM7, TK1, UBE2C).
- **Prevention of circularity:** when a programme is compared with the proliferation score or adjusted for proliferation, the 20 proliferation markers contained in that programme are first removed from the programme score. For the mitotic cell cycle set (the positive control), the adjusted result is not interpreted.

- **Age:** the midpoint of GTEx's released age bands (20–29, ..., 70–79), i.e. 25, 35, ..., 75.
- **Technical covariates:** RIN, ischaemic time, $\log_{10}$ library size. Fields not present in the recount3 metadata will not be used, and that fact will be recorded.
- **Biological covariates:** sex, Hardy scale of death circumstances (missing values as a separate category).
- **Sample exclusions:** samples not belonging to the four tissue types above; missing age; the lowest 1% of library size within a tissue type. Where one donor has several samples of one tissue type, the one with the largest library size.
- **Definition of the correlation:** the primary measure is the **partial Spearman ρ** after regressing the technical covariates out of both variables (GTEx is post-mortem and surgical material, so technical variation is large). The raw Spearman ρ is a sensitivity analysis.
- **Age model:** $\text{score} \sim \text{age} + \text{sex} + \text{Hardy} + \text{RIN} + \text{ischaemic time} + \log_{10} \text{ library size}$; the proliferation-adjusted model adds the proliferation score to this. $\text{Attenuation} = 1 - \beta_{\text{adj}}/\beta$.

4. Primary hypotheses and decision criteria

	Hypothesis	Success	Failure
H1 culture replication	Partial ρ between the splicing-96 score and the proliferation score in GTEx cultured fibroblasts	$\rho \geq 0.5$	$\rho < 0.3$ (0.3–0.5 is partial replication)
H2 coupling weaker in tissue	The splicing-96–proliferation ρ in lower-leg skin tissue is smaller than in cultured fibroblasts	Fisher z difference, one-sided $P < 0.05$	no difference, or the opposite
H3 post-mitotic tissue boundary	Partial ρ between splicing-96 and proliferation in skeletal muscle	$\rho < 0.3$	$\rho \geq 0.5$
H4 attenuation of the age effect in culture	In cultured fibroblasts, the age effect on splicing-96 is	attenuation $\geq 50\%$	attenuation $< 25\%$. If the unadjusted age $P \geq 0.05$, report as

	Hypothesis	Success	Failure
	attenuated after adjustment for proliferation		not evaluable
H5 collagen dissociation in culture	Partial ρ between collagen formation and proliferation in cultured fibroblasts	$ \rho < 0.3$	$ \rho \geq 0.5$
H6 positive control for tissue ageing	Collagen formation falls with age in lower-leg skin tissue	$\beta < 0, P < 0.05$	$\beta \geq 0$ or $P \geq 0.05$ (in which case the whole tissue age analysis is interpreted as underpowered)

Note on H2: because the epidermis of skin contains proliferating basal keratinocytes, the coupling may persist in tissue and H2 may fail. In that case the conclusion is restricted to “skin tissue is not a post-mitotic tissue”, and the judgement about the boundary is left to H3.

5. Exploratory analyses (no success criteria)

- **S1** Compare the partial ρ with proliferation of the 24 programmes side by side in culture, the two skin sites and skeletal muscle (the tissue version of Fig. 8A). [Translator’s note: “Fig. 8A” is the panel number in the draft current at the time of writing; it is Fig. 6a in the revised manuscript.]
- **S2** Compare the age slope of collagen formation between sun-exposed (lower leg) and non-exposed (suprapubic) skin.
- **S3** Donor-level concordance of programme scores between a donor’s cultured fibroblasts and skin tissue.
- **S4** Skin tissue composition sensitivity: add keratinocyte marker (KRT5, KRT14, KRT1, KRT10) and dermal fibroblast marker (COL1A1, COL1A2, DCN, LUM, PDGFRA) scores as covariates.
- **S5** Full table of age effects for the 24 programmes $\times$ tissue types, before and after adjustment for proliferation.

6. Multiple testing

- H1–H6: success is decided by the effect-size criteria above. The P value of each hypothesis is reported together after Benjamini–Hochberg correction within the six primary hypotheses.
- S1, S5: within each tissue type, the proliferation family and the age family are BH-corrected separately (as in the existing audit).

7. Reporting principles

- Report both successes and failures. Failed hypotheses will not be removed from the manuscript.
- Do not change definitions, thresholds or sample-exclusion rules after seeing results. If a change is unavoidable, record the date and reason in the amendment history below and report **the result under the original definition as well**.
- The material analysed is reprocessed public data and no causal claim is made. GTEx age is given in ten-year bands, so age effects are interpreted only at band resolution.

Amendment history

(none)

Outcomes

Recorded in results_gtex_culture_tissue_boundary_ko.md. Final samples after the preregistered exclusions: cultured fibroblasts 511 donors (492 with complete technical covariates), lower-leg skin 739, suprapubic skin 626, skeletal muscle 839; 388 donors contributed both cultured fibroblasts and lower-leg skin.

	Preregistered criterion	Estimate obtained	<i>P</i>	BH-FDR	Verdict
H1	$\rho \geq 0.5$ succeeds; $\rho < 0.3$ fails	partial $\rho = +0.739$ in cultured fibroblasts ($n = 492$)	1.2×10^{-85}	7.1×10^{-85}	supported
H2	one-sided $P < 0.05$ for the Fisher z difference	lower-leg skin $+0.225$ ($n = 739$) versus culture $+0.739$; $z = -11.95$	3.2×10^{-33} (one-sided)	9.6×10^{-33}	supported
H3	$\rho < 0.3$ succeeds; $\rho \geq 0.5$	partial $\rho = -0.115$ in	8.5×10^{-4}	1.7×10^{-3}	supported

	Preregistered criterion	Estimate obtained	<i>P</i>	BH-FDR	Verdict
	fails	skeletal muscle (<i>n</i> = 839)			
H4	attenuation $\geq 50\%$ succeeds; not evaluable if the unadjusted age <i>P</i> ≥ 0.05	unadjusted age effect -0.025 per decade	0.33	0.33	not evaluable
H5	$ \rho < 0.3$ succeeds; $ \rho \geq 0.5$ fails	partial $\rho = +0.049$ for collagen formation in culture	0.28	0.33	supported
H6	$\beta < 0$ and <i>P</i> < 0.05 succeeds	collagen formation in lower-leg skin $+0.027$ per decade, i.e. rising	0.018	0.027	not supported (opposite direction)

Because H6 failed, the preregistered rule was applied: age effects in the GTEx tissues are treated as not validated by a positive control and are not used to support any claim in the manuscript. This applies to the tissue columns of S5 and to the S2 comparison of sun-exposed with non-exposed skin. It was noted at the time, without changing that rule, that the failure was a significant effect in the opposite direction rather than an absence of power.

Deviations and post hoc additions

In the order in which they occurred. The preregistration's own amendment history is empty; the first two items below are recorded in the outcome record, and the rest are later additions, each labelled post hoc.

1. **Before the analysis was run — the P value attached to H4.** The preregistration did not state which P value belonged to H4. The draft script attached the proliferation-adjusted age P ; before any result was seen this was changed to the **unadjusted** age P , which is the quantity the preregistered rule uses to decide whether H4 is evaluable. The decision rule itself was unchanged. Recorded in `results_gtex_culture_tissue_boundary_ko.md`.
2. **Before the analysis was run — a failed first execution.** The first run stopped before R started because the output directory did not exist. The directory was created and the same script was run again. No definition changed.
3. **The splicing set is the 96-gene version, fixed by the preregistration.** After the preregistration was locked, a fourth donor cohort (GSE226189) was removed from the study on 2026-09-13 because its processing batches coincide with age strata, which changed the study-wide splicing set from 96 genes to 177. The preregistered verdicts above stand on the 96-gene file, which was not rewritten. Repeating the same analysis with the 177-gene set is a **post hoc sensitivity analysis**: cultured fibroblasts +0.766, lower-leg skin +0.203, skeletal muscle −0.130, and the culture age effect −0.024 per decade unadjusted ($P = 0.35$) and −0.052 adjusted ($P = 0.001$). Every verdict is the same and H4 remains not evaluable.
4. **Post hoc — 32 off-trend cultures.** After the outcomes were known, 32 of the 511 cultures were found to lie off the main trend at low proliferation; they are enriched in particular RNA-extraction and sequencing batches and have lower RIN. Excluding them raises the raw correlation from 0.72 to 0.89. The preregistered estimate includes them, and the exclusion is reported only as an observation.
5. **Post hoc — range restriction and variance-matched subsampling** (2026-09-15, `results_gtex_posthoc_ko.md`, `public_data_tierA/derived/gtex_posthoc/`). Added in response to the objection that proliferation varies less between tissue donors than between cultures. The regression slope, which range restriction does not affect, falls from 0.554 in culture to 0.167 in lower-leg skin, 0.072 in suprapubic skin and −0.127 in muscle; a Thorndike case-II correction predicts $\rho = 0.653$ for lower-leg skin against the 0.225 observed; and cultures subsampled to the skin's proliferation spread

give a median partial ρ of 0.872 (95% range 0.834–0.910) with a median-centred window and 0.770 (0.701–0.857) with the tails down-weighted rather than discarded.

6. **Post hoc — adjustment for cell composition** (2026-09-15). This extends the preregistered exploratory analysis S4 in three ways, all of them after the outcomes were known: the marker panels were enlarged (keratinocyte KRT5, KRT14, KRT1, KRT10, TP63, IVL, LOR; fibroblast COL1A1, COL1A2, DCN, LUM, PDGFRA, **PDGFRB**; immune PTPRC; and for muscle myofibre ACTA1, MYH1, MYH2, MYH7, CKM and satellite-cell PAX7, MYF5); the fibroblast panel thus keeps the five genes the preregistration named, PDGFRA among them, and **adds PDGFRB**; and the adjustment was applied to the splicing-proliferation coupling itself, whereas S4 as written applied it to age effects. With composition in the covariates the partial ρ between the splicing-96 score and the proliferation score falls from +0.225 ($P = 6.6 \times 10^{-10}$) to +0.101 ($P = 0.006$) in lower-leg skin ($n = 739$ donors), from +0.056 ($P = 0.17$) to -0.058 ($P = 0.15$) in suprapubic skin ($n = 626$), and from -0.115 ($P = 8.5 \times 10^{-4}$) to -0.040 ($P = 0.25$) in muscle ($n = 839$), while culture is essentially unchanged (+0.739 to +0.722, $n = 492$; `public_data_tierA/derived/gtex_posthoc/composition_adjusted.tsv`). The fibroblast panel was revised once, on the same day and after the first post hoc run: that run had PDGFRB in place of the preregistered PDGFRA and gave +0.111 ($P = 0.003$) in lower-leg skin, -0.052 ($P = 0.19$) in suprapubic skin, -0.021 ($P = 0.55$) in muscle and +0.721 in culture, after which PDGFRA was restored and PDGFRB kept as an addition, with no other change to markers, covariates, samples or model; the values reported above and in the main text are from the revised panel, and the first-run table is kept as `composition_adjusted_first_definition.tsv`. The preregistered S4 result on the age effect, which was run with the preregistered panel, stands as recorded (lower-leg skin splicing-96 age effect -0.032 , $P = 0.037$, becoming -0.013 , $P = 0.32$, with keratinocyte and fibroblast markers), and is not used as evidence, since it is a tissue age effect.
7. **Post hoc — genotype-driven positive control for donor concordance** (2026-09-15). The preregistered exploratory analysis S3 found no donor-level concordance between cultured fibroblasts and skin for any of the 25 programme scores ($\rho = -0.10$ to +0.13, none at $FDR < 0.05$). To establish that the design can detect donor identity at all, genes whose expression is strongly genotype-dependent were tested in the same 388 donors and the same matrices: *GSTM1* $\rho = +0.750$ ($P = 3 \times 10^{-71}$) and *ERAP2* $\rho = +0.749$ ($P = 5 \times 10^{-71}$), neither explained by sex, together with seven sex-chromosome genes at $\rho = 0.666$ to 0.736. This control was not preregistered.

8. **Post hoc — redefinition of the comparison cohort** (2026-09-15). Fig. 8c compares the GTEx cultures with the fibroblast donor cohort GSE113957. That cohort was redefined after the outcomes were known, on grounds independent of the GTEx analysis: ten progeria donors (nine of them in the 142-sample analysed set) and 26 donors under 20 years of age were removed, and cell repository, sequencing instrument and sex were added as covariates, leaving 107 normal adults. For the preregistered 96-gene set this changes the comparison cohort's estimates from -0.093 per decade unadjusted and -0.031 adjusted, as recorded at the time, to -0.128 unadjusted ($P = 4 \times 10^{-6}$) and -0.033 adjusted (95% CI -0.064 to -0.002 , $P = 0.04$). Nothing in the GTEx analysis itself was recomputed.
9. **Post hoc — combined residual age effect** (2026-09-16, `public_data_tierA/derived/residual_meta/residual_age_effect_meta.tsv`). The proliferation-adjusted age effects on the preregistered 96-gene set in the GTEx cultures (-0.053 per decade, $n = 492$ donors) and in the 107-donor cohort (-0.033) were combined by inverse variance: -0.042 per decade (95% CI -0.065 to -0.019 , 599 donors). Unadjusted estimates were not combined. This was not preregistered; it uses the cultures only, not the tissues to which the rule that followed the failure of H6 applies. Withdrawn at revision (2026-09-18): the pooled value is no longer reported anywhere in the manuscript or its tables; the two estimates are shown side by side in Fig. 8c and Supplementary Table 5 and are not combined, because one is an attenuation and the other a suppression effect in cultures with no unadjusted age effect, and with two studies the between-study variance is not estimable.
10. **Post hoc — specificity of the coupling in GTEx cultures** (2026-09-16, `public_data_tierA/derived/machinery_specificity/gtex_culture_partials.tsv`; Supplementary Fig. S5b, Supplementary Table 5). The comparison of splicing definitions with translation, Metabolism of RNA and rRNA processing, made first against the counted division rate, was repeated against the 20-marker proliferation score in the 511 GTEx cultures. The asymmetry seen against the counted division rate was not reproduced: with the other score held constant, translation keeps a partial ρ of 0.51 to 0.59 and the splicing definitions 0.28 to 0.41. This was not preregistered, and it uses the cultures only.

Two further discrepancies are recorded for completeness. The preregistration estimated the download at about 277 MB; the files actually retrieved total 265 MB, with SHA-256 digests in `public_data_tierA/gtex/gtex_sha256.txt`. And the panel numbering

referred to in S1 is that of the draft current on 2026-09-13, not of the revised manuscript.

Supplementary Note 2. Preregistration for the methylation clock analysis

What this document is

The methylation analysis reported in Supplementary Fig. S7 was planned before the data were seen. The plan below was written in Korean on 16 September 2026, and its SHA-256 digest was recorded before the methylation matrix of GSE179847 was downloaded; the analysis script `scripts/revision/run_methylation_clock.R` recomputes the digest and stops if it does not match. The digest is

be51a7550c982e2071ebdb80f36ab7098965351c6e14a65434e8df275505d929

and it still matches the current file, which has not been amended since it was locked.

As with Supplementary Note 1, this was an internal record and was not deposited with a public registry, so a reader cannot verify independently when it was written; what can be verified is that the document the analysis ran against is the document reproduced here, and that its criteria are the criteria against which the outcomes below are reported, including the three the analysis failed. The translation is complete rather than summarised.

Translation

Preregistration: the relationship between methylation clocks and the transcriptomic proliferation readout

This document was written **before downloading** GSE179847 (Cellular Lifespan Study DNA methylation, EPIC array, 479 samples). The SHA-256 is recorded, then the data are downloaded, and the analysis script checks the digest before it runs. After locking, additions are made only to the amendment history at the end of the document.

1. Background and the question

This study has shown that transcriptional programmes in cultured fibroblasts, splicing factors among them, are readouts of the **measured division rate**. That methylation clocks run fast in culture (Sturm 2019) and that cell division causes methylation loss in late-replicating domains

(Endicott 2022) are already known. **“The clock follows division too” is therefore not itself a new question.**

The new question is one. **In the same cultures, are the two biomarker families — the transcriptomic proliferation readout and the methylation clock — measuring the same variable?** GSE179847 and GSE179848 come from the same cell lines and the same passages, so they can be matched directly at the sample level.

2. Data

- **GSE179847:** GSE179847_Cell_lifespan_DNA_m_processed_matrix.csv.gz (about 2.0 GB). Rows are CpGs, columns are per-sample beta values and detection *P* values. Sample identifiers are sentrix IDs, linked to GSM and unique_variable_name through the GEO sample list.
- **GSE179848:** the RNA-seq data already held. 328 libraries carry a measured division rate.
- **Clock coefficients:** the Horvath 2013 multi-tissue clock (353 CpGs). The skin and blood clock (391) and PhenoAge (513) are used only if their coefficient tables can be obtained from a public source; if they cannot, that fact is recorded.

3. Definitions fixed in advance

- **DNA_m age:** computed with each clock’s published coefficients. For Horvath 2013 the published inverse transformation (adult basis) is used as it stands. Missing CpGs are replaced by the cohort mean of that CpG and **the number replaced is reported**.
- **Clock acceleration:** the residual of DNA_m age regressed on days in culture within a cell line.
- **Measured division rate:** 24 / population_doubling_time_uhours_per_division (divisions per day). This name is not used for any other dataset.
- **Transcriptomic proliferation score:** the mean *z* of the existing 20 markers (MKI67 and the rest).
- **Splicing score:** the mean *z* of the existing 177-gene pre-mRNA processing set.
- **Matching:** only samples with exactly the same unique_variable_name are treated as a pair. If no exact match exists, matching is restricted to samples identical in cell line, treatment, oxygen and passage, and the number matched is reported.
- **Note on normalisation:** this analysis does not reproduce the original paper’s pipeline (BMIQ and the rest), so **absolute age estimates are approximations**. Absolute values are not interpreted; only correlations and residuals are.

4. Hypotheses and success criteria

	Hypothesis	Success criterion
M1 (sanity check)	In untreated healthy cell lines, DNAm age increases with cumulative divisions	$\rho \geq 0.3, P < 0.05$
M2	Clock acceleration is associated with the measured division rate	$ \rho \geq 0.3, P < 0.05$
M3 (the main test)	In the same cultures, the transcriptomic proliferation score is associated with DNAm age	$ \rho \geq 0.3, P < 0.05$
M4	The association between the splicing score and DNAm age shrinks when the measured division rate is adjusted for	the partial correlation falls by at least 50%

- If M1 fails, the clock computation is taken to be wrong and **M2-M4 are not interpreted**. In that case the failure is recorded as it stands.
- If any hypothesis comes out in the opposite direction it is reported as it stands. Criteria are not changed after results are seen.
- Multiple testing is corrected across the four hypotheses with Benjamini-Hochberg.

5. Exclusion criteria

- Measurements with a detection P value above 0.01 are treated as missing.
- Samples missing 30% or more of the clock CpGs are excluded and their number reported.
- Samples with no division rate, or a rate of zero or below, are excluded (the same rule as the RNA analysis).

6. Reporting principles

- That absolute DNAm ages are approximations is stated in the figure and in the text.
- This analysis occupies **no more than one panel and one paragraph** in the paper. The claim is limited to one thing: whether the two biomarker families read the same variable in culture.
- Endicott 2022 and Sturm 2019 are named as prior work, and what this analysis re-confirms is distinguished from what it adds.

7. Amendment history

(none)

Outcomes

The analysis was run on 16 September 2026 with scripts/revision/run_methylation_clock.R, which verified the digest above before it started. Of 479 arrays, none was missing more than 30% of clock CpGs; the detection filter set 0.018% of measurements to missing and the 99 remaining gaps were filled with the cohort mean of the CpG. Clock CpGs available: 334 of 353 (Horvath 2013), 391 of 391 (skin and blood), 513 of 513 (PhenoAge), 63 of 69 (Hannum). After the division-rate exclusion, 471 methylation samples from nine cell lines and nineteen treatments remained, and exact matching on unique_variable_name gave 109 cultures with both a clock and transcriptional scores, from lines HC1, HC2 and HC4.

	Test	ρ	P	BH-FDR	n	Criterion met
M1	DNAm age vs cumulative doubling times, untreated healthy lines	+0.356	8×10^{-4}	0.0015	86	yes
M2	clock acceleration vs measured division rate	-0.014	0.76	0.76	471	no
M3	proliferation score vs DNAm age, paired cultures	+0.162	0.093	0.124	109	no
M4	splicing	-0.338	3×10^{-4}	0.0013	109	no

Test	ρ	P	BH-FDR	n	Criterion met
vs DNAm age with division rate held constant					

M1 was met, so M2–M4 are interpreted under the preregistered rule. M2 and M3 failed on the effect-size criterion fixed in advance ($|\rho| \geq 0.3$), not on significance alone: the 95% confidence interval for M3 is -0.03 to $+0.34$, so a correlation of the preregistered size is not excluded by these 109 cultures, and a larger paired set could change the verdict. M4 asked for attenuation of at least 50%; the association did not attenuate but changed sign, from $+0.15$ unadjusted to -0.34 with division rate held constant (an attenuation of -131%), so the criterion is not met. That sign change is recorded and not interpreted.

Supporting numbers, none of them preregistered tests. Within the 109 paired cultures the transcriptional scores follow the measured division rate as they do throughout the study (splicing $\rho = +0.77$, proliferation $+0.69$, and the two agree with each other at $+0.82$), while DNAm age follows it at $+0.41$ before time in culture is removed and at zero after. Three further clocks, computed the same way, give acceleration-rate correlations of $+0.18$ for skin and blood ($P = 7 \times 10^{-5}$), $+0.14$ for Hannum ($P = 0.003$) and $+0.06$ for PhenoAge ($P = 0.18$); all are below the preregistered 0.3.

Deviations and post hoc additions

1. **The Hannum clock was added.** The preregistration named Horvath 2013, the skin and blood clock and PhenoAge. Hannum's clock was computed as well, because its coefficient table was obtained at the same time. It is reported only as a supporting comparison in Supplementary Fig. S7d and carries none of the preregistered verdicts.
2. **How “untreated healthy cell lines” was operationalised for M1.** The preregistration did not define the phrase. It was taken as the six healthy-donor lines (HC1–HC6) under the study's own Control treatment label at 21% oxygen, which gives $n = 86$. The SURF1 patient lines and every treated or hypoxic culture are excluded from M1 only; M2 uses all 471.
3. **Clock acceleration was fitted with a common time slope.** “The residual of DNAm age regressed on days in culture within a cell line” was implemented as a single regression on days with a cell-line term, which gives every line the same slope. Fitting a separate slope per

line instead gives $\rho = +0.092$ against the measured division rate ($P = 0.045$), still well below the preregistered 0.3, and the correlation within individual lines ranges from -0.74 to $+0.40$. The verdict on M2 is the same either way.

4. **Post hoc — the confidence interval reported for M3.** The preregistration fixed a threshold but did not ask for an interval. The Fisher-z interval is given in the manuscript and above so that the failure is not read as evidence of absence.
5. **Reporting limit** (recorded 2026-09-17). The preregistration allowed one panel and one paragraph. The analysis is reported as one four-panel supplementary figure (Supplementary Fig. S7: the sanity check M1, the tests M2 and M3, and the supporting clocks; M4 is not drawn, and its estimate is in Supplementary Table 8) so that every outcome is visible, with one Results paragraph, one sentence of Discussion, one Methods subsection and Supplementary Table 8. The claim remains limited as preregistered, and no claim about clocks is made.

Supplementary Data

Machine-readable tables accompany this file.

File	Deposited as	Contents
Supplementary Data 1	SupplementaryData1_contrasts.tsv	The 63 contrasts: change in the 177-gene splicing set, in the cell-cycle set and in the splicing set without the 23 shared genes, with the class of each contrast (leave_out_defining_series.tsv in the same folder gives the refit without the three set-defining series)
Supplementary Data 2	SupplementaryData2_gtex_programmes_by_tissue.tsv	Programme scores against proliferation and donor age in every GTEx tissue, with age effects before and after adjustment
Supplementary Data 3	SupplementaryData3_gene_level_by_cohort.ts	For each of the four cohort definitions: the

File	Deposited as	Contents
	v	proliferation-age correlation, the machinery-proliferation correlation, and the number of genes with an age association before and after adjustment for proliferation
Supplementary Data 4	SupplementaryData4_programme_division_rate.tsv	Division-rate correlations for every programme, with and without cell-cycle genes
Supplementary Data 5	SupplementaryData5_residual_specification.tsv	The residual age effect under every model specification — splicing definition, comparator programme and covariate set (Supplementary Table 5)
Supplementary Data 6	SupplementaryData6_composition_adjusted.tsv	Partial correlation with proliferation before and after cell-composition marker scores, for six sets in four tissues (Fig. 8d, Supplementary Fig. S8b)
Supplementary Data 7	SupplementaryData7_library_scores_vs_counted_rate.tsv	Counted division rate, 20-marker proliferation score and the 177- and 96-gene splicing scores for the 328 counted libraries (Figs 2 and 7a)
Supplementary Data 8	SupplementaryData8_intervention_prediction.tsv	Out-of-sample prediction of the intervention classes: contrasts, series, median change and

File	Deposited as	Contents
		residual under each prediction scheme (Supplementary Fig. S4, Supplementary Results 5)
Supplementary Data 9	SupplementaryData9_senescence_panels.tsv	The seven senescence panels: gene counts, cell-cycle content, correlation with the counted division rate with and without cell-cycle genes, and the age effect before and after adjustment (Supplementary Fig. S6, Supplementary Table 7)
Supplementary Data 10	SupplementaryData10_methylation_tests.tsv	The four preregistered methylation clock tests with their correlations, <i>P</i> values, BH-FDR and verdicts (Supplementary Table 8)
Supplementary Data 11	SupplementaryData11_gene_vs_counted_rate.tsv	Every expressed gene of the counted resource with its correlation to the counted division rate, its mean expression, and whether it belongs to the 177-gene set or to the twenty named factors; for the 177, the fold change per division per day with its <i>P</i> and FDR (Fig. 2c)
Supplementary Data 12	SupplementaryData12_junction_psi_vs_proliferation.tsv	Every testable splicing event of the donor cohort with its inclusion level and its association with the proliferation score and

File	Deposited as	Contents
Supplementary Data 13	SupplementaryData13_gene_age_effects.tsv	with donor age, before and after adjustment (Fig. 3a, e) Each of the 177 genes in the donor cohort: the age effect in $\log_2$ counts per million per decade with its <i>P</i> and FDR, before and after adjustment for proliferation
Supplementary Data 14	SupplementaryData14_classified_splicing_events.tsv	Every event of Supplementary Data 12 classified against GENCODE v41: event type, the alternative exon or segment with its transcript-type annotation, the junction measured, its share in the slowest and fastest thirds of donors, and the tests against proliferation and age (Fig. 3b-d)